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Fast Simulation of Neutrino Oscillations in Matter

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Abstract

This Master’s Thesis presents **tpeanuts**, an open-source PyTorch framework for fast and differentiable neutrino-oscillation calculations. Starting from a validated translation of the PEANUTS numerical core, **tpeanuts** reformulates three-flavour oscillations, matter effects and solar-neutrino propagation in terms of batched tensor operations. This enables parallel evaluation over energies, trajectories and oscillation parameters on CPU or GPU, while retaining automatic differentiation through the physical calculation.

The framework extends the original solar-neutrino scope of PEANUTS to atmospheric propagation through the Earth, interchangeable solar and terrestrial density models, and beyond-the-Standard-Model scenarios including non-standard interactions and sterile-neutrino mixing. Its modular structure separates neutrino sources, propagation media and new-physics scenarios from detector and inference components. A forward-folding detector layer converts propagated fluxes into experimental observables, and a gradient-based statistical layer uses the resulting differentiable models to infer oscillation and detector parameters from published data.

The translated numerical core was validated against the original PEANUTS implementation and independently against **nuSQuIDS**, with additional CPU/GPU benchmarks used to assess the computational behaviour of the tensor formulation. The complete framework was then applied to solar, atmospheric and reactor-neutrino calculations and to detector-level analyses of experimental data. In the most complete case, a fit to the Daya Bay near/far reactor-antineutrino data determines $\sin^2 2\theta_{13}$ and Δm_{31}^2 jointly within 1–2% of the collaboration’s published values.

These results establish **tpeanuts** as a common differentiable framework in which neutrino propagation, detector response and statistical inference can be treated within the same computational model, while preserving the physical conventions of the original PEANUTS implementation.

Keywords

neutrino oscillations; PEANUTS; PyTorch; differentiable programming; GPU computing; open-source software; MSW effect; solar neutrinos; atmospheric neutrinos; reactor antineutrinos; statistical inference.

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1 Introduction

Neutrino oscillations are one of the clearest experimental indications that the Standard Model of particle physics must be extended, as established by the atmospheric-neutrino measurements of Super-Kamiokande and the solar-neutrino measurements of the Sudbury Neutrino Observatory (Ahmad et al., 2002; Fukuda et al., 1998). A neutrino produced with a definite flavour, such as an electron, muon or tau neutrino, propagates as a coherent superposition of mass eigenstates. The interference among these components gives rise to flavour transitions whose probabilities depend on the neutrino energy, the travelled distance, the mixing parameters and, when propagation occurs in matter, the electron density of the medium (Giunti & Kim, 2007; Mikheyev & Smirnov, 1985; Wolfenstein, 1978).

Modern neutrino analyses evaluate these probabilities repeatedly, inside flux predictions, detector simulations, likelihood scans and global fits, so the numerical implementation directly affects the feasibility of high-statistics studies and large parameter scans. Existing tools such as PEANUTS (Gonzalo & Lucente, 2024) provide compact routines for neutrino propagation, matter mixing and solar-neutrino probabilities, but array-based implementations are not always optimal for current workflows involving GPUs, TPUs, automatic differentiation and batched execution.

This thesis documents the development of `tpeanuts`, a PyTorch (Paszke et al., 2019) reproduction and extension of the PEANUTS neutrino-oscillation toolkit. Its numerical basis is a validated translation of the PEANUTS core, including three-flavour propagation in vacuum and matter and solar-neutrino probabilities. The translation preserves the physical conventions of the original code while expressing the calculations as tensor operations. It thereby supports parallel evaluation of many oscillation configurations, execution on either CPU or GPU, and differentiation of the resulting probabilities for parameter estimation.

The translated core is extended through modules for atmospheric-neutrino propagation in the Earth, alternative solar and terrestrial matter-density models and beyond the Standard Model such as non-standard interactions and sterile neutrinos. A detector-level forward-folding and statistical-inference layer connects these calculations to published data from several experiments. The oscillation engine and its extensions are distributed as a public, open-source Python package with installation and usage documentation (section 5.6), allowing the calculations presented here to be inspected, reused and extended.

1.1 Motivation

The main motivation for `tpeanuts` is the need for fast, repeated neutrino-oscillation evaluation, both for large-scale simulation (atmospheric-flux studies, detector-response studies, BSM scans) and, ultimately, for fitting real data: a gradient-based fit to the published spectrum of a detector requires the oscillation probability, the detector response and the likelihood to be evaluated, and differentiated, at every optimiser step. PyTorch provides a natural environment for this because it combines vectorised tensor operations, GPU support and automatic differentiation, letting a single implementation serve fast forward simulation and gradient-based inference alike.

The original PEANUTS implementation is tied to a specific solar model and a fixed treatment of

matter effects, which limits its use where the sensitivity of a result to these choices must be tested. Building **tpeanuts** on a flexible tensor framework makes it possible to add such options, including alternative solar and Earth density models, perturbative treatments of the matter potential, and new-physics scenarios such as non-standard interactions (NSI) and sterile neutrinos, in a modular way, without compromising the validated core of the oscillation formalism. This same modularity is what makes it practical to add a detector-response layer (target composition, energy resolution, backgrounds, efficiencies) and an inference layer (likelihood construction, gradient-based fitting, uncertainty estimation) on top of the oscillation engine, and to reuse both across experiments as different as a reactor (Daya Bay), a liquid-scintillator solar detector (Borexino), a large-volume atmospheric detector (IceCube DeepCore) and a heavy-water Cherenkov detector (SNO).

The intended use of **tpeanuts** is not restricted to the analyses in this thesis. Its common differentiable interface covers solar, atmospheric and BSM calculations that would otherwise require several purpose-built programs. The package is available on GitHub together with installation instructions and companion documentation for its individual modules.

Validation is carried out at several levels. Independent numerical strategies for the same propagation problem are first compared within their stated approximations. Intermediate quantities and final probabilities are then checked against PEANUTS and **nuSQuIDS** (Argüelles et al., 2022). Finally, complete calculations are compared with established phenomenological and experimental results, including the solar day–night asymmetry and detector data.

1.2 Objectives

The general objective of this Master’s Thesis is to develop and document **tpeanuts** as a modular, extensible and publicly available PyTorch framework for neutrino-oscillation phenomenology, built on a validated translation of the core of PEANUTS and extended into a detector-level statistical-inference framework validated against real experimental data. The specific objectives are:

- to review the three-flavour oscillation formalism used by PEANUTS, including vacuum propagation and matter effects, and to translate its numerical core to PyTorch tensors while preserving physical conventions, units and sign choices;
- to design a modular architecture in which vacuum/solar/Earth/atmosphere models, matter and perturbative treatments, and new-physics scenarios (NSI, sterile neutrinos) can be selected and reused consistently across the application;
- to define a validation strategy comparing **tpeanuts** against PEANUTS and **nuSQuIDS** at the level of intermediate quantities and final probabilities, complemented by the reproduction of well-known theoretical and experimental results and by a CPU/GPU benchmarking methodology;
- to implement a detector-agnostic forward-folding layer (cross sections, target composition, energy response, efficiency, backgrounds) and to specialise it for real experiments;
- to implement a gradient-based statistical-inference layer (Poisson and Gaussian likelihoods, LBFGS fitting, Laplace/Fisher uncertainty estimation, likelihood scans) built on PyTorch autograd;
- to validate the full detector-inference chain against real, published data, with Daya Bay as

the primary case study, and to document its application to Borexino, IceCube DeepCore and SNO;

- to release `tpeanuts` publicly, as a documented, installable, open-source package intended for reuse by the neutrino-phenomenology community, rather than as a closed thesis-specific codebase.

1.3 Document structure

Section 2 introduces the theoretical framework: neutrino masses, mixing, oscillations, matter effects and the statistical-inference formalism. Section 3 explains how the numerical core was translated and validated, and how the modular and BSM extensions and the inference chain were assessed. Section 4 describes the software implementation, including the detector and inference layers. Section 5 reports the comparisons with PEANUTS and `nuSQuIDS`, the atmospheric, density-model, NSI and sterile-neutrino applications, and the analyses of experimental data, with Daya Bay as the main case study. Section 6 gives the conclusions and identifies future work.

2 Theoretical Framework

This section summarises the physical ingredients required by `tpeanuts`, focusing on the conventions that must be preserved when updating PEANUTS to a tensor-based implementation; it is not a complete review of neutrino physics.

2.1 Neutrino propagation fundamentals

This subsection collects the formalism common to every propagation regime considered in this thesis: the definition of the mixing matrix, vacuum oscillations, and the modification of the propagation Hamiltonian by ordinary matter. The specific neutrino sources and new-physics extensions built on top of this common core are treated separately in the following subsections.

2.1.1 Neutrino mixing

One of the fundamental discoveries in particle physics is that neutrinos are produced and detected in flavour eigenstates, whereas they propagate through space as mass eigenstates. Since these two bases do not coincide, a neutrino created with a definite flavour evolves into a quantum superposition of different flavours during propagation, giving rise to the phenomenon of neutrino oscillations (Giunti & Kim, 2007).

Within the Standard Model extended to include neutrino masses, the flavour eigenstates $|\nu_\alpha\rangle$, $\alpha = e, \mu, \tau$, are related to the mass eigenstates $|\nu_i\rangle$, $i = 1, 2, 3$, through a unitary transformation known as the Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrix (Maki et al., 1962; Pontecorvo, 1957, 1958),

$$|\nu_\alpha\rangle = \sum_{i=1}^3 U_{\alpha i}^* |\nu_i\rangle, \quad (1)$$

or, equivalently,

$$|\nu_i\rangle = \sum_{\alpha=e,\mu,\tau} U_{\alpha i} |\nu_\alpha\rangle. \quad (2)$$

The matrix U is unitary, $U^\dagger U = U U^\dagger = \mathbb{I}$, which guarantees the conservation of probability during neutrino propagation.

Physical interpretation. The matrix elements $U_{\alpha i}$ represent the probability amplitudes that a neutrino with flavour α is found in the mass eigenstate i ; consequently, $|U_{\alpha i}|^2$ gives the probability that a neutrino produced as flavour α propagates in the mass eigenstate i .

Standard parametrisation. For three active Dirac neutrinos, the PMNS matrix is written as the product of three elementary rotations, $U_{\text{PMNS}} = R_{23}(\theta_{23}) U_{13}(\theta_{13}, \delta_{\text{CP}}) R_{12}(\theta_{12})$, yielding the standard Particle Data Group (PDG) parametrisation (Workman et al., 2022)

$$U_{\text{PMNS}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta_{\text{CP}}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta_{\text{CP}}} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta_{\text{CP}}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta_{\text{CP}}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta_{\text{CP}}} & c_{23}c_{13} \end{pmatrix}, \quad (3)$$

with $c_{ij} = \cos \theta_{ij}$, $s_{ij} = \sin \theta_{ij}$. Each rotation has a clear physical meaning: θ_{12} governs mixing between the first and second mass eigenstates and dominates solar-neutrino oscillations together with the smaller mass splitting Δm_{21}^2 ; θ_{23} controls mixing between the second and third eigenstates and is primarily responsible for atmospheric-neutrino oscillations; θ_{13} couples the first and third eigenstates and, although considerably smaller than the other mixing angles, plays a crucial role in matter effects and enables sensitivity to CP violation. The complex phase δ_{CP} introduces a difference between neutrino and antineutrino oscillation probabilities, providing a possible source of CP violation in the lepton sector.

Oscillation parameters. Oscillation experiments probe differences between squared neutrino masses, $\Delta m_{ij}^2 = m_i^2 - m_j^2$; the absolute masses themselves do not enter the oscillation probability. Only two independent mass-squared differences exist, Δm_{21}^2 and Δm_{31}^2 (or equivalently Δm_{32}^2), together with the three mixing angles and the CP-violating phase,

$$\{\theta_{12}, \theta_{13}, \theta_{23}, \Delta m_{21}^2, \Delta m_{31}^2, \delta_{\text{CP}}\}. \quad (4)$$

These six quantities completely determine neutrino oscillation probabilities within the standard three-neutrino oscillation framework, with current best-fit values obtained through global analyses combining solar, atmospheric, reactor and accelerator data. The NuFIT collaboration's NuFIT 5.2 release (Esteban et al., 2020) fixes the oscillation-parameter presets used throughout this work's forward-simulation and validation notebooks; the more recent NuFIT 6.1 update (Esteban et al., 2025), based on the same underlying analysis (Esteban et al., 2024), is used separately, only as an external benchmark in the real-data detector-level fits of section 3.4, not as a simulation input.

2.1.2 Vacuum oscillations

Once produced in a flavour eigenstate, neutrinos propagate in vacuum as coherent superpositions of mass eigenstates. Since each mass eigenstate accumulates a different quantum phase, their interference is the physical origin of neutrino oscillations (Giunti & Kim, 2007). The evolution is

governed by the Schrödinger equation,

$$i \frac{d}{dt} |\nu(t)\rangle = H |\nu(t)\rangle, \quad (5)$$

where H is the effective Hamiltonian describing neutrino propagation.

Vacuum Hamiltonian. In vacuum, H is diagonal in the mass basis, $H_{\text{mass}} = \text{diag}(E_1, E_2, E_3)$, with $E_i = \sqrt{p^2 + m_i^2}$. Since neutrinos produced in laboratory, solar and atmospheric experiments are highly relativistic, $m_i^2 \ll p^2$, the energy can be expanded as $E_i \simeq p + m_i^2/(2E)$, with $E \simeq p$ the neutrino energy. The common momentum term contributes only an overall global phase, e^{-ipL} , which has no observable consequences and can be dropped, leaving the effective vacuum Hamiltonian

$$H_{\text{vac}} = \frac{1}{2E} \text{diag}(m_1^2, m_2^2, m_3^2). \quad (6)$$

Flavour evolution requires transforming H_{vac} into the flavour basis through the PMNS matrix,

$$H_f = U H_{\text{vac}} U^\dagger = \frac{1}{2E} U \text{diag}(m_i^2) U^\dagger. \quad (7)$$

Evolution operator. For a constant Hamiltonian, the Schrödinger equation is solved by $|\nu(L)\rangle = e^{-iHL} |\nu(0)\rangle$. For a neutrino produced in flavour α , $|\nu(0)\rangle = |\nu_\alpha\rangle$, the transition amplitude to flavour β after a distance L is

$$A_{\alpha \rightarrow \beta}(L, E) = \sum_{i=1}^3 U_{\beta i} \exp\left(-i \frac{m_i^2 L}{2E}\right) U_{\alpha i}^*, \quad (8)$$

with oscillation probability $P_{\alpha \rightarrow \beta} = |A_{\alpha \rightarrow \beta}|^2$. Unitarity of the PMNS matrix guarantees $\sum_\beta P_{\alpha \rightarrow \beta} = 1$, i.e. conservation of probability.

Oscillation phases. Oscillations arise from the phase differences accumulated by the mass eigenstates, controlled by the mass-squared differences $\Delta m_{ij}^2 = m_i^2 - m_j^2$, which define the phase entering the transition amplitude of equation (8), $\Delta_{ij} = \Delta m_{ij}^2 L / 2E$. Squaring a coherent two-state superposition introduces a half-angle: the two-flavour reduction of equation (8) gives $P = 1 - \sin^2(2\theta) \sin^2(\Delta_{ij}/2)$, so it is $\Delta_{ij}/2$, not Δ_{ij} itself, that governs the observable probability. In the practical units customary in oscillation phenomenology (Workman et al., 2022), $\Delta_{ij}/2 = 1.267 \Delta m_{ij}^2 [\text{eV}^2] L [\text{km}] / E [\text{GeV}]$; oscillations are thus governed by the ratio L/E rather than by the baseline or the energy separately.

2.1.3 Matter effects

The vacuum formalism above assumes that neutrinos propagate through empty space. In many settings relevant to **tpeanuts**, such as the solar interior, the Earth's mantle and core, or a supernova envelope, neutrinos instead traverse dense media, where coherent forward scattering off the particles of the medium modifies the propagation Hamiltonian. This is the matter effect, also known as the Mikheyev–Smirnov–Wolfenstein (MSW) effect (Giunti & Kim, 2007; Mikheyev & Smirnov, 1985; Wolfenstein, 1978). Because coherent forward scattering transfers essentially no momentum to the medium, it does not remove neutrinos from the beam; instead, it modifies

the phase accumulated during propagation, which is described as an additional effective potential in the Schrödinger equation.

Matter potential. Neutrinos interact with matter through both neutral-current (NC) and charged-current (CC) weak processes. NC interactions, $\nu_\alpha + f \rightarrow \nu_\alpha + f$ with f an electron, proton or neutron, affect all active flavours equally; since oscillation probabilities depend only on relative phases between flavours, this common contribution is unobservable and is omitted from the oscillation Hamiltonian. Electron neutrinos additionally undergo the CC process $\nu_e + e^- \rightarrow \nu_e + e^-$, mediated by $W^\pm$ exchange, which introduces a flavour-dependent potential (Wolfenstein, 1978)

$$V_{\text{CC}} = \sqrt{2} G_F n_e, \quad (9)$$

where G_F is the Fermi coupling constant and n_e the local electron number density. This additional, electron-flavour-specific interaction breaks the flavour symmetry of vacuum propagation and is the origin of the MSW effect.

Effective Hamiltonian. The full Hamiltonian governing flavour evolution in matter is obtained by adding the matter potential to the vacuum Hamiltonian (Giunti & Kim, 2007; Wolfenstein, 1978),

$$H = H_{\text{vac}} + V_{\text{mat}}, \quad H_{\text{vac}} = \frac{1}{2E} U \text{diag}(m_1^2, m_2^2, m_3^2) U^\dagger, \quad V_{\text{mat}} = \sqrt{2} G_F n_e \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (10)$$

where V_{mat} is diagonal in the flavour basis, since only the electron flavour experiences the CC interaction. For antineutrinos, both the matter potential changes sign, $V_{\text{mat}} \rightarrow -V_{\text{mat}}$, and the mixing matrix is complex-conjugated, $U \rightarrow U^*$ (equivalently $\delta_{\text{CP}} \rightarrow -\delta_{\text{CP}}$), a convention detailed in section 4.2.2. In matter, the propagation basis becomes density-dependent, $H = U_m \text{diag}(\lambda_1, \lambda_2, \lambda_3) U_m^\dagger$, with U_m the effective mixing matrix and λ_i the effective eigenvalues; as the electron density varies along the trajectory, both evolve continuously, responsible for the characteristic energy dependence of solar-neutrino oscillations.

2.1.4 Numerical vs. perturbative propagation

Once the electron density varies along the trajectory, the constant-density solution above no longer applies exactly, and `tpeanuts` offers two independent strategies for the resulting Schrödinger equation, a segmented numerical evolver and a first-order perturbative one.

First-order perturbative method. Within a segment over which the density fluctuates around a mean value $\bar{n}_e$, the matter Hamiltonian is split as $n_e(x) = \bar{n}_e + \delta n_e(x)$, $H_f(x) = H_0 + H_1(x)$, with $H_0 = H_{\text{kin}} + \bar{V}_{\text{CC}} P_e$ the constant part and $H_1(x) = \delta V_{\text{CC}}(x) P_e = \sqrt{2} G_F \delta n_e(x) P_e$ the residual. Diagonalising the traceless part of H_0 gives eigenvalues λ_a and spectral projectors M_a , so the constant-density piece is exact,

$$S_0(L) = e^{-iH_0L} = \sum_a e^{-i(\lambda_a + \text{Tr } H_0/N)L} M_a, \quad (11)$$

and the density fluctuation contributes the first Dyson term (Akhmedov et al., 2007),

$$S_1(L) = -i \int_0^L S_0(L-x) H_1(x) S_0(x) dx = -i \sum_{a,b} I_{ab} M_a P_e M_b, \quad (12)$$

where I_{ab} is the analytic oscillatory integral of $\delta n_e(x) e^{-i(\lambda_b - \lambda_a)x}$, evaluated in closed form for the polynomial density fits used for the Earth and atmosphere (section 3.2). The segment evolver is $S \simeq S_0 + S_1$, and full-trajectory propagation composes these segment operators in order. Being accurate only to first order in δn_e , this truncated evolver need not be exactly unitary; an optional reunitarization step restores it explicitly.

Segmented numerical method. The numerical evolver instead divides the trajectory into N_{step} intervals and exponentiates the Hamiltonian sampled at each interval’s midpoint,

$$H_j = H_f(x_{j+1/2}), \quad S_j = \exp(-iH_j \Delta x_j), \quad S_{\text{num}} = S_{N-1} \cdots S_1 S_0. \quad (13)$$

Every factor is exactly unitary, being the exponential of a Hermitian sampled Hamiltonian; the only approximation is the piecewise-constant replacement of the continuous density profile, whose ordered product converges to the exact path-ordered exponential as $\max_j \Delta x_j \rightarrow 0$. This composition of constant-density layers is a standard treatment of neutrino propagation through matter of varying density (Fishbane & Kaus, 2001); unlike the perturbative method, it does not expand in δn_e and is not amenable to reunitarization.

The perturbative evolver is faster (section 5.1.1) but only first-order accurate in the density fluctuation. The numerical evolver is exact in that sense, up to a step-size-dependent discretisation error, at a higher computational cost.

2.2 Neutrino sources

The oscillation formalism introduced above is universal: the same Schrödinger equation governs flavour evolution regardless of how the neutrino is produced. What varies between sources is the physical environment of production and, consequently, the density profile traversed before detection, which determines the relative importance of vacuum oscillations, matter effects and decoherence. Each source is therefore sensitive to a different region of the oscillation parameter space.

The current version of **tpeanuts** is designed primarily for the simulation of solar and atmospheric neutrinos, which together probe a broad range of energies, baselines and matter densities. Owing to its modular architecture, however, the propagation engine itself is independent of the neutrino source, so that additional sources can be incorporated with minimal modifications, as discussed for reactor and accelerator neutrinos below.

2.2.1 Solar neutrinos

Solar neutrinos are produced by thermonuclear fusion in the solar core, where temperatures reach $\sim 1.5 \times 10^7$ K and densities exceed 150 g cm^{-3} (Vinyoles et al., 2017). The dominant production

mechanisms are the proton–proton (pp) chain and the CNO cycle, generating neutrinos from a few hundred keV up to ~ 20 MeV (Vinyoles et al., 2017).

Unlike reactor or accelerator neutrinos, solar neutrinos are produced deep inside a medium whose density decreases continuously from the core to the photosphere, so that flavour evolution is strongly affected by the MSW effect. For the Large Mixing Angle (LMA) solution favoured by current oscillation data (Esteban et al., 2020), propagation through the Sun is highly adiabatic over most of the relevant energy range: the neutrino remains in the same instantaneous matter eigenstate as the local density changes (Haxton, 1986). Within this approximation, the flavour state produced at radius r is projected onto the local matter eigenstates, which evolve adiabatically into vacuum mass eigenstates as the neutrino exits the Sun. After propagating over the Earth–Sun distance (1 AU), the mass eigenstates lose mutual coherence due to wave-packet separation and the finite energy resolution of detectors (Giunti & Kim, 2007), so the flavour transition probabilities reduce to an incoherent sum,

$$P_{\nu_e \rightarrow \nu_\alpha}^{(f)}(E) = \sum_{i=1}^3 |U_{\alpha i}|^2 T_i^{(f)}(E), \quad (14)$$

where $T_i^{(f)}(E)$ is the probability that a neutrino produced through solar reaction f emerges from the Sun as vacuum mass eigenstate ν_i . Since production is spatially distributed throughout the solar interior, $T_i^{(f)}$ is obtained by averaging over the corresponding radial production profile,

$$T_i^{(f)}(E) = \int_0^{R_\odot} w_f(r) P_{\nu_e \rightarrow \nu_i}(E, r) dr, \quad (15)$$

with $w_f(r)$ the normalised production distribution predicted by the Standard Solar Model (SSM). The solar module in **tpeanuts** uses radial electron-density profiles together with production distributions for the principal channels (pp , pep , hep , ${}^7\text{Be}$, ${}^8\text{B}$, ${}^{13}\text{N}$, ${}^{15}\text{O}$, ${}^{17}\text{F}$), using standard solar models such as B16(GS98) and B16(AGSS09met) (Vinyoles et al., 2017). Optional non-adiabatic corrections based on the Landau–Zener formalism (Parke, 1986) are also available for the regions where the adiabatic approximation becomes less accurate. Because solar neutrinos experience both matter-enhanced conversion inside the Sun and, at night, regeneration inside the Earth, they are one of the most powerful probes of neutrino propagation in matter and provide stringent constraints on the solar mixing parameters and on possible non-standard interactions.

2.2.2 Atmospheric neutrinos

Atmospheric neutrinos are produced when high-energy cosmic rays interact with nuclei in the atmosphere, generating hadronic cascades dominated by pions and kaons; the decay of these mesons and of the resulting secondary muons produces both muon and electron neutrinos, spanning energies from a few hundred MeV to beyond the TeV scale (Honda et al., 2007). Unlike solar neutrinos, they are produced over a wide range of altitudes and arrive at terrestrial detectors from all directions, with baselines extending from ~ 20 km for downward-going events to nearly the Earth’s diameter ($\sim 12\,742$ km) for upward-going neutrinos that cross the planet.

This broad coverage in energy and baseline makes atmospheric neutrinos sensitive to several oscillation regimes at once: downward-going events are dominated by atmospheric or vacuum

oscillations, while upward-going neutrinos experience significant matter effects while traversing the Earth’s mantle and core. The interplay between the two regimes produces the characteristic oscillation patterns in the $(E, \cos \theta_z)$ plane known as oscillograms. Within **tpeanuts**, atmospheric-neutrino calculations combine the three-flavour propagation formalism with realistic Earth density profiles, such as the Preliminary Reference Earth Model (PREM) (Dziewonski & Anderson, 1981), and differential atmospheric flux tables from the Honda parametrisation (Honda et al., 2007) or from cascade solvers such as MCEq (Fedynitch et al., 2015).

2.2.3 Reactor antineutrinos

Nuclear reactors are intense, well-characterised sources of electron antineutrinos, emitted in the beta decays of the neutron-rich fission fragments of ^{235}U , ^{238}U , ^{239}Pu and ^{241}Pu , with a spectrum extending up to ~ 8 MeV and a rate set by the reactor’s thermal power. Because flavour evolution is entirely determined by the vacuum Hamiltonian of section 2.1, and reactor baselines ($\lesssim 2$ km for a near/far experiment such as Daya Bay) traverse negligible Earth matter, reactor antineutrinos follow the coherent three-flavour vacuum survival probability, with θ_{13} governing $\bar{\nu}_e \rightarrow \bar{\nu}_e$ disappearance together with the atmospheric-scale splitting. At this baseline $\sin^2 \Delta_{31}$ and $\sin^2 \Delta_{32}$ are numerically close enough, differing only through the small Δ_{21} phase, that what the disappearance channel directly resolves is the effective, ordering-insensitive combination $\Delta m_{ee}^2 = \cos^2 \theta_{12} \Delta m_{31}^2 + \sin^2 \theta_{12} \Delta m_{32}^2$, rather than Δm_{31}^2 itself, which is only recovered after fixing θ_{12} and Δm_{21}^2 externally (section 5.5.1). **tpeanuts** implements a full reactor-antineutrino observable chain on top of this probability, from the per-isotope Huber–Mueller emission spectrum (Huber, 2011; Mueller et al., 2011) through the inverse-beta-decay cross section (Vogel & Beacom, 1999) to a detector-response model folding the true antineutrino spectrum into an observable prompt-energy spectrum, described in section 4.1.1 and validated against real Daya Bay data in section 5.5.1.

2.2.4 Other neutrino sources

Other sources, such as accelerator-based long-baseline beams, require only specifying the appropriate initial flavour composition, energy spectrum, baseline and matter-density profile within the same propagation engine, without any change to the oscillation formalism itself.

2.3 New physics

Beyond-the-Standard-Model scenarios can be incorporated into the same formalism through modifications of the effective Hamiltonian and, in the case of sterile neutrinos, of the PMNS matrix itself. Two such extensions are considered in **tpeanuts**: non-standard interactions and sterile neutrinos.

2.3.1 Non-standard interactions (NSI)

Although the three-flavour framework introduced above explains the vast majority of neutrino oscillation observations, it is widely regarded as an effective low-energy description rather than a complete theory: the Standard Model does not account for the origin of neutrino masses, nor does it explain the nature of dark matter, the matter–antimatter asymmetry or the hierarchy of fermion masses. Many theories beyond the Standard Model (BSM) therefore predict additional neutrino interactions that would manifest as small deviations from the standard oscillation

paradigm.

Among the most studied such scenarios are non-standard interactions (NSI), in which neutrinos experience additional weak-scale interactions with matter mediated by new heavy particles (Grossman, 1995). At energies well below the mediator mass scale, these effects can be described with an effective field theory in which the heavy degrees of freedom are integrated out and encoded in dimensionless effective couplings, providing a model-independent framework for a broad class of BSM scenarios without specifying the underlying microscopic theory (Farzan & Tortola, 2018).

NSI can in general affect neutrino production, detection and propagation; this work considers only *propagation* NSI, in which new neutral-current-like interactions modify the coherent forward scattering of neutrinos through matter. Since coherent forward scattering only changes the propagation phase, these interactions appear as an additional contribution to the matter Hamiltonian,

$$H_{\text{mat}} = \sqrt{2}G_F n_e \left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} \varepsilon_{ee} & \varepsilon_{e\mu} & \varepsilon_{e\tau} \\ \varepsilon_{e\mu}^* & \varepsilon_{\mu\mu} & \varepsilon_{\mu\tau} \\ \varepsilon_{e\tau}^* & \varepsilon_{\mu\tau}^* & \varepsilon_{\tau\tau} \end{pmatrix} \right], \quad (16)$$

where the dimensionless couplings $\varepsilon_{\alpha\beta}$, normalised to the electron density n_e for a given medium composition, quantify the interaction strength relative to the Standard Model weak interaction. Only the traceless part of the diagonal is observable because a common shift to all three entries produces an unobservable global phase. The convention used in `tpeanuts` is therefore $\varepsilon_{\mu\mu} = 0$. The off-diagonal terms introduce flavour-changing coherent interactions with no Standard Model analogue. For antineutrinos, $\varepsilon_{\alpha\beta}$ is complex-conjugated together with the sign flip described in section 4.2.2.

NSI couple separately to the matter fermions e , u and d . Using electrical neutrality, $n_p = n_e$, their contribution can be written as $\varepsilon_{\alpha\beta}(x) = \varepsilon_{\alpha\beta}^{(e+p)} + (n_n(x)/n_e(x))\varepsilon_{\alpha\beta}^n$. These two effective terms are implemented separately, with $\varepsilon^n = 0$ by default to recover equation (16). The distinction is relevant because the neutron-to-electron ratio differs between the Sun and the Earth and enters the LMA-Dark degeneracy discussed below. NSI leave $H_{\text{vac}} = \frac{1}{2E}UM^2U^\dagger$ unchanged and modify only the matter term in $H = H_{\text{vac}} + H_{\text{mat}}$, so their effect depends on density, composition and energy.

Phenomenologically, NSI modify the effective mixing angles and mass splittings in matter, shift the MSW resonance, alter the day–night regeneration of solar neutrinos, and can introduce degeneracies with the standard oscillation parameters. A well-known example is the LMA-Dark solution (Esteban et al., 2018; Miranda et al., 2006), where a large negative ε_{ee} compensates a transformation of the solar mixing angle, yielding oscillation probabilities nearly indistinguishable from the standard Large Mixing Angle (LMA) solution; such degeneracies illustrate the value of combining different classes of neutrino experiments to disentangle standard oscillation physics from possible BSM effects. Current data constrain most individual NSI couplings to be subdominant with respect to the Standard Model weak interaction, but not universally: near degeneracies such as LMA-Dark, large couplings survive precisely because they are compensated by a parameter transformation rather than excluded outright, and couplings involving the τ flavour remain

only weakly bounded on their own; global analyses combining oscillation, scattering and collider measurements give increasingly stringent, but data-set-dependent, limits (Biggio et al., 2009; Esteban et al., 2018).

In the implementation, NSI alter only the matter contribution to the Hamiltonian, so the propagation algorithms used for the standard three-flavour model remain unchanged. A general complex Hermitian NSI matrix represents both flavour-conserving and flavour-changing couplings. Standard-Model and BSM predictions can therefore be evaluated with the same PyTorch propagation routines.

2.3.2 Sterile neutrinos

The Standard Model contains three active flavours, ν_e , ν_μ and ν_τ , all of which participate in the weak interaction through charged- and neutral-current processes. Several BSM models, together with a number of experimental anomalies reported over the past decades, have nonetheless motivated the existence of additional neutrino states that are singlets under the Standard Model gauge group. Because they do not couple directly to the weak interaction, these hypothetical states are known as *sterile* neutrinos (Böser et al., 2020). Although they cannot be produced or detected through Standard Model weak processes, sterile neutrinos may mix with the active flavours through the neutrino mass matrix and thus participate in flavour oscillations as additional propagation eigenstates, modifying the observed oscillation pattern without introducing any new interaction at the detector.

The simplest and most widely studied extension is the $3 + 1$ model (Giunti & Lasserre, 2019; Kopp et al., 2013), in which a single sterile state is added to the three active flavours. The flavour basis becomes $(\nu_e, \nu_\mu, \nu_\tau, \nu_s)$ and the propagation basis four mass eigenstates $(\nu_1, \nu_2, \nu_3, \nu_4)$, enlarging the 3×3 PMNS matrix into a 4×4 unitary mixing matrix, built here as

$$U_4 = R_{23} \Delta R_{24}(\theta_{24}, \delta_{24}) R_{34}(\theta_{34}, \delta_{34}) R_{13} \Delta^\dagger R_{12} R_{14}(\theta_{14}, \delta_{14}), \quad (17)$$

extending the three-flavour factorisation of section 2.1, with R_{23} , Δ , R_{13} , R_{12} embedded unchanged and three new active-sterile rotations R_{14} , R_{24} , R_{34} , each carrying its own CP phase. This ordering is **tpeanuts**'s own implementation convention rather than a universal standard, and is not equivalent to a block-embedded U_{PMNS} followed by the sterile rotations. Beyond the three-flavour parameters, the $3 + 1$ framework introduces three new mixing angles θ_{14} , θ_{24} , θ_{34} ; **tpeanuts** parametrises the sterile sector with three raw phases δ_{14} , δ_{24} , δ_{34} , of which only two are physically independent, consistent with the general $N \times N$ count of $(N - 1)(N - 2)/2$ physical phases (three for $N = 4$, including δ_{CP}). The extension also adds a fourth mass eigenstate m_4 and $\Delta m_{41}^2 = m_4^2 - m_1^2$, typically $\gg |\Delta m_{31}^2|$ (of order 1 eV^2) in short-baseline-anomaly phenomenology (Kopp et al., 2013), giving oscillation lengths much shorter than the atmospheric or solar ones.

Unlike NSI, which modify only the matter contribution, sterile neutrinos change the formalism more fundamentally: the vacuum Hamiltonian itself must be enlarged. The formalism of section 2.1 is expressed in natural units, $\hbar = c = 1$; **tpeanuts**'s internal normalisation rescales it by a purely numerical reference length L_{scale} , together with the explicit $\hbar c$ conversion needed once L_{scale} is expressed in metres rather than natural (inverse-energy) units, without changing the underlying physics. Consistently with the matter potentials below, the enlarged vacuum term is

$$H_{\text{vac}} = \frac{L_{\text{scale}}}{2E\hbar c} U_4 \text{diag}(m_1^2, m_2^2, m_3^2, m_4^2) U_4^\dagger, \quad (18)$$

so the new eigenstate contributes its own phase, $\exp(-im_4^2 L/2E)$ in natural units, to the evolution operator. Matter effects are modified too. The sterile state feels neither the charged- nor the neutral-current (NC) potential common to the three active flavours, $V_{\text{NC}} = \mp \frac{1}{\sqrt{2}} G_F n_n L_{\text{scale}}$ (from the neutron density n_n). V_{NC} is an unobservable global phase in the pure three-flavour case, but an observable relative potential once this NC-blind sterile state is added after subtracting the common phase,

$$H_{\text{mat}} = \text{diag}(V_{\text{CC}}, 0, 0, -V_{\text{NC}}), \quad (19)$$

with $V_{\text{CC}} = \sqrt{2} G_F n_e L_{\text{scale}}$. This NC term is an optional addition, active whenever a neutron-density profile is supplied; omitting it recovers the CC-only sterile matter term used as the default here, an approximation once θ_{24} or θ_{34} are non-zero, and one not to be assumed negligible in general: the NC potential is standard practice for matter-affected active-sterile propagation in the literature, underlying the Earth-matter resonance exploited by atmospheric and long-baseline ν_μ - ν_s searches (IceCube Collaboration, Abbasi, et al., 2020). Either way, the modified matter Hamiltonian alters the effective eigenvalues and can generate additional resonant conversion in dense media.

Phenomenologically, sterile neutrinos may produce anomalous flavour disappearance, unexpected appearance channels, distortions of oscillation spectra and modified matter-induced effects, and have been extensively investigated in connection with the LSND and MiniBooNE anomalies, the reactor antineutrino anomaly and the Gallium anomaly, although no conclusive evidence has yet established their existence (Böser et al., 2020). Cosmological observations independently constrain light sterile neutrinos through their impact on Big Bang nucleosynthesis, the effective number of relativistic degrees of freedom, and large-scale structure formation.

The oscillation engine accepts Hamiltonians of arbitrary dimension. In the 3+1 implementation, the 4×4 mixing matrix, enlarged mass spectrum and matter Hamiltonian are therefore handled by the propagation algorithms also used for the three-flavour model. The present implementation covers the $3 + 1$ scenario; models with additional sterile states would require further model definitions, but not a different propagation scheme.

2.4 Statistical inference formalism

Comparing an oscillation prediction against real, binned experimental data requires a statistical framework connecting predicted and observed counts, a rule for minimising the resulting statistic with respect to the oscillation and detector parameters, and a way of estimating the uncertainty on the fitted values. `tpeanuts` implements this framework directly on top of PyTorch’s automatic differentiation, so that the same gradient used to build the oscillation Hamiltonian also drives the fit.

For a set of bins with predicted counts $N_k(\theta)$ and observed counts n_k^{obs} , the Poisson (Baker–Cousins) likelihood-ratio statistic (Baker & Cousins, 1984)

$$-2 \ln L_{\text{Poisson}} = 2 \sum_k \left[N_k(\theta) - n_k^{\text{obs}} + n_k^{\text{obs}} \ln \frac{n_k^{\text{obs}}}{N_k(\theta)} \right] \quad (20)$$

is used for absolute event counts, and vanishes exactly when $N_k = n_k^{\text{obs}}$ bin by bin. For derived observables that are not themselves Poisson-distributed, such as a ratio of counts, a Gaussian statistic with propagated uncertainty is used instead, $-2 \ln L_{\chi^2} = \sum_k [(R_k(\theta) - R_k^{\text{obs}})/\sigma_k]^2$. Externally measured nuisance parameters (background normalisations, energy-scale pulls) are constrained by an additional Gaussian penalty term on θ itself, added to either statistic.

The best-fit point $\hat{\theta} = \arg \min_{\theta} [-2 \ln L(\theta)]$ is found with the quasi-Newton LBFGS optimiser, using the exact gradient of the total statistic obtained by backpropagating through the full physics chain, oscillation probability, detector response and likelihood, rather than finite differences. The uncertainty on $\hat{\theta}$ is estimated through the Laplace approximation, in which the Fisher information at the optimum, $\mathcal{I} = \frac{1}{2} \nabla_{\theta}^2 (-2 \ln L)$, obtained as the autograd Hessian of the statistic, gives the parameter covariance $\Sigma = \mathcal{I}^{-1}$, with $\sigma_j = \sqrt{\Sigma_{jj}}$. Where the local Laplace approximation is inadequate, in particular along directions the data barely constrain, a direct two-dimensional scan of $-2 \ln L$ over a parameter grid, optionally profiling the remaining free parameters at each grid point, provides confidence contours at Wilks'-theorem asymptotic thresholds, valid under standard regularity conditions but liable to fail near physical boundaries, degeneracies or low statistics; since **tpeanuts** performs no Monte Carlo (e.g. Feldman–Cousins) coverage calibration, these contours should be read as asymptotic, not empirically calibrated.

2.5 Related computational tools

tpeanuts interfaces with several public tools for cross-validation (**nuSQuIDS** Argüelles et al., 2022) or as external physical inputs (MCEq Fedynitch et al., 2015 and the Honda/HKKM tables Honda et al., 2007 for atmospheric flux, **pymis** Emmert et al., 2021 for atmospheric density, CORSIKA Heck et al., 1998 exploratorily); the corresponding software modules are described in section 4.1.

3 Methodology

The development begins with the translation of the PEANUTS core and the numerical and performance comparisons described in section 3.1. The resulting implementation is extended to a modular propagation framework and to beyond-the-Standard-Model scenarios in sections 3.2 and 3.3. The source–medium–detector–inference calculation is subsequently tested with real data in section 3.4. Packaging, documentation and public release of the code form the final stage described in section 3.5.

Figure 1 groups the extensions with respect to PEANUTS into three areas. The computational changes comprise GPU execution, batching and autograd; the physics modules add atmospheric neutrinos, Earth propagation, NSI and sterile neutrinos; and the data-analysis components provide detector response, likelihoods, gradient-based inference and real-data fits. All three areas are included in the public, documented software release.

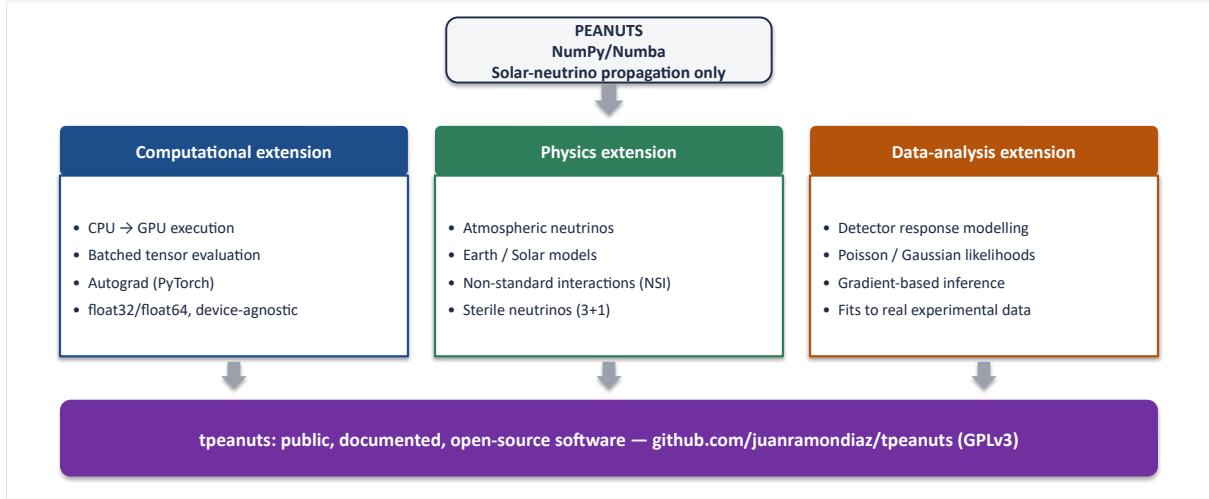


Figure 1: From PEANUTS to **tpeanuts**: the three independent extensions developed in this thesis.

3.1 Core translation: methodology, validation and benchmarking

3.1.1 Translation methodology

The development of **tpeanuts** follows a systematic translation process: each PEANUTS (Gonzalo & Lucente, 2024) module (PMNS construction, vacuum and matter Hamiltonians, matter mixing, numerical propagation, solar production and Earth propagation, detector exposure) is first interpreted from the underlying theoretical formalism of section 2.1, separating the physical algorithm from implementation-specific detail, and then reformulated using native, batched PyTorch (Paszke et al., 2019) tensor operations, so that large grids of energies, baselines and production points are evaluated within a single vectorised call. PEANUTS’s physical conventions, the PMNS phase, the neutrino/antineutrino sign, mass-eigenstate ordering, matter-potential normalisation, are explicitly verified at every step: intermediate quantities (mixing matrices, Hamiltonians, eigenvalues, matter potentials, oscillation phases) are compared directly against the original implementation before validating the final oscillation probabilities, since sign or unit-conversion errors can otherwise produce numerically stable but physically incorrect results.

Central to this translation is the batched, autograd-preserving PyTorch tensor formulation detailed architecturally in sections 4.1 and 4.3: every quantity carries an explicit batch dimension evaluated in a single vectorised call, with the fast-forward routines behind most results in sections 5.3 and 5.4 disabling autograd for speed while the inference models of section 3.4 build on the same differentiable primitives directly.

3.1.2 Validation strategy

Two types of comparison are used.

- **Intrinsic Validation.** For the internal comparison, the perturbative/analytical and purely numerical matrix-exponential evolvers of section 2.1 solve the same propagation problem independently and are compared wherever both apply.
- **Extrinsic Validation.** The external comparison uses two reference implementations and examines the calculation at three levels:

- low-level quantities, such as PMNS matrix elements, kinetic phases and matter

potentials;

- intermediate physical outputs, such as effective mixing angles and eigenvalues in matter and mass-eigenstate weights for solar neutrinos;
- final oscillation probabilities, energy- and production-averaged and, where a detector model exists, folded into predicted event rates.

Because PEANUTS implements only solar-neutrino propagation, its comparison covers vacuum propagation, the solar MSW effect, production-weighted survival probabilities and Earth regeneration. `nuSQuIDS` (Argüelles et al., 2022) provides an independent calculation for the solar and atmospheric regimes using a different numerical method. The differences are examined in connection with the conventions and approximations used for production averaging and loss of coherence. Probability conservation, positivity and the unitarity of the evolution operator are checked for the configurations included in these comparisons.

3.1.3 Benchmarking

In addition to physical correctness, evaluating computational performance is an important objective, both **intrinsically** (CPU vs GPU, float32 vs float64, numerical versus perturbative, etc.) and **extrinsically** (against PEANUTS and `nuSQuIDS`), since efficient evaluation of large numbers of configurations is a principal motivation for the tensor-based design.

Intrinsic Benchmarking. Batched GPU evaluation distributes the calculation of section 3.1 over many arithmetic units. Kernel dispatch introduces an approximately fixed overhead, whereas the computation time grows with the amount of work. Sequential execution can therefore be faster for small batches, while larger batches make progressively better use of GPU parallelism, as tested for the perturbative/numerical comparison in section 5.1.1 and against the reference codes in section 5.1.2. Tensors are allocated on the target device before timing, GPU operations are explicitly synchronised, and warm-up iterations are excluded. The reported values are averages over repeated measurements on CPU and GPU in single and double precision; double precision is also used for comparisons with the reference codes.

Extrinsic Benchmarking. Wherever the calculations are directly comparable, particularly for vacuum and atmospheric propagation in `nuSQuIDS` and for vacuum and earth propagation in PEANUTS, the three packages use matched oscillation parameters, density profiles, precision and physical scenarios. The benchmarks vary the number of energy points, baseline or density samples, batch size, device, precision and oscillation scenario. Performance is reported as wall-clock speed-up relative to PEANUTS and `nuSQuIDS`. The comparison reflects their different evaluation models: `nuSQuIDS` uses one solver object per energy and angle point, PEANUTS vectorises NumPy operations over a trajectory, and `tpeanuts` evaluates the complete grid as a tensor batch. The dependence of the measured speed-up on grid size is presented in section 5.1.2.

3.2 Extension to a modular architecture: sources, media and detectors

The validated core of section 3.1.2 is deliberately decoupled from any specific neutrino source, propagation medium or detector: the same Hamiltonian and evolution machinery serve solar, atmospheric and reactor sources; vacuum, solar and Earth matter media, with interchangeable density models; and the detector models of table 2 built on top of the resulting probabilities. The same engine could serve accelerator sources with no change to the propagation core. This

separation is what allows the framework to extend beyond PEANUTS’s original solar-only scope, to atmospheric-neutrino propagation through the Earth, alternative solar and Earth density models, and new detectors, without touching the propagation core already validated above.

3.3 BSM extensions: NSI and sterile neutrinos

Beyond-the-Standard-Model scenarios are incorporated as modifications of the same Hamiltonian (sections 2.3.1 and 2.3.2), validated by a signature specific to each extension. For **NSI**, the benchmark is the LMA-Dark degeneracy (Esteban et al., 2018; Miranda et al., 2006): the simultaneous transformation $\theta_{12} \rightarrow \pi/2 - \theta_{12}$, $\varepsilon_{ee} \rightarrow -(2 + \varepsilon_{ee})$ must reproduce a day–night–asymmetry sign flip consistent with the near-degenerate solution, exercising the sign dependence of the implementation. For **sterile neutrinos**, three presets (a null-mixing reference, a global-fit-motivated best fit, and a short-baseline-anomaly benchmark) are checked against analytic expectations: exact recovery of the Standard-Model limit at zero sterile mixing and unitarity of the enlarged 4×4 mixing matrix to machine precision.

3.4 Inference methodology: the full source–medium–detector chain

Validating the detector and inference layers exercises the full chain built in the sections above end to end, source flux, medium propagation, detector response and statistical inference, against real, published experimental data, since no independent reference implementation exists to compare against. Three complementary checks are applied, in full where the detector model is complete and in part otherwise, across the implemented experiments:

- a **closure test**, a prediction-reproduction check against real published data, evaluating the full forward model at the experiment’s own published best-fit parameters.
- an **absolute-normalisation audit**, when the closure test reveals a systematic offset, tracing it through every stage of the forward-model chain against independently published values to identify and quantify its dominant sources;
- a **parameter-recovery fit**, minimising the appropriate likelihood of section 2.4 with the quasi-Newton LBFGS optimiser (Liu & Nocedal, 1989), comparing the fitted values against the experiment’s own published best fit and against the global-fit value from NuFIT 6.1 (Esteban et al., 2024, 2025).

This methodology is applied in full to Daya Bay, the primary case study of section 5.5.1, and in summary form, with a parameter-recovery fit, to IceCube DeepCore, SNO phase I and SNO phase II. For Borexino, a direct fit of the oscillation parameters to the collaboration’s own published $P_{ee}(E)$ points succeeds, while the full detector-level event-rate chain remains at the implementation/unit-test stage (section 5.5.2).

3.5 Open-source release: a public, documented software package

Releasing **tpeanuts** as public, open-source software was a deliberate part of this methodology of work. The full source code, together with the notebooks used for every validation, benchmark and analysis in this thesis, is publicly available on GitHub under the GNU General Public License v3 (GPLv3), the same licence as the original PEANUTS implementation it builds on, with an editable, development-mode install (`pip install -e .`) and a documented layout. Alongside the code itself, seven companion $\text{\LaTeX}$ documents provide module-by-module architectural and

physics detail beyond the scope of this thesis, each with its own figures and, for the inference documents, its own real-data results. Full details, including the repository URL and the notebook directory tree, are given in section 5.6.

4 Technical Development

`tpeanuts` takes PEANUTS as its starting point, but structures the resulting code in a modular, hierarchical way that favours extending it with new features and physical models, rather than reproducing PEANUTS's own flat structure. This section describes the main components of that structure.

4.1 Software architecture

`tpeanuts` reimplements the PEANUTS NumPy/Numba neutrino-oscillation and flux-propagation toolkit in PyTorch and adds detector-level statistical inference. The implementation represents Hamiltonians, evolution operators, oscillation probabilities, fluxes, predicted event counts and likelihoods as functions of broadcastable tensors. Within the available memory, this formulation evaluates grids of energies, angles and detector configurations in one vectorised call instead of invoking the calculation separately for every point.

The same physics primitives support PyTorch automatic differentiation. Torch records the differentiable tensor operations in the calculation and uses that graph to evaluate derivatives with respect to the selected inputs, up to floating-point precision and subject to the differentiability of each operation. Autograd is disabled in the forward-simulation routines used for most physics results, but enabled in the experiment-specific models of section 4.1.1. Their gradients can consequently be propagated from the likelihood through the detector and oscillation calculations in the fits of section 3.4. This tensor-based, differentiable interface differs from the legacy implementation and from the C++/pybind11 interface used here for `nuSQuIDS` (Arguelles et al., 2022), which is invoked with one solver object for each energy and angle point.

`tpeanuts` therefore operates in two complementary modes:

- **Fast-forward simulation:** dedicated pipelines assemble the propagation primitives into complete, physically meaningful observables for each production environment, a solar-neutrino pipeline and an atmospheric-neutrino pipeline being the two main instances, evaluated with autograd disabled for speed.
- **Inference:** the same primitives are reused with autograd enabled, so that gradients propagate through the full chain, from the free oscillation parameters down to the predicted observable, letting a gradient-based optimiser fit those parameters directly against real, published experimental results.

Figure 2 sketches the resulting three-tier decomposition, application, domain and core layers, detailed below. This hierarchy is itself supported by a set of blocks that sit outside it: a configuration block centralising model choices, a shared utility layer providing standard device/precision management and generic numerical primitives, and a collection of external-tool integrations, third-party packages such as MCEq and external data repositories such as the Honda/HKKM flux tables, consumed as configuration input to the physics above.

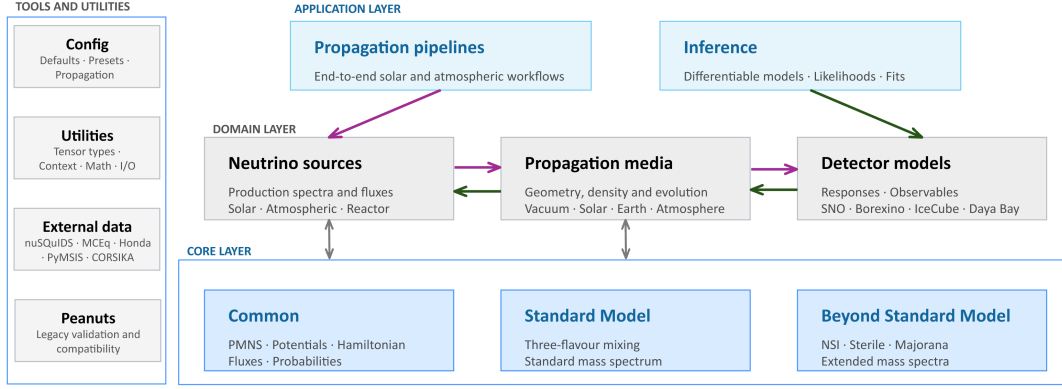


Figure 2: Modular architecture of `tpeanuts`: application, domain and core layers, supported by configuration, utility, external-data and legacy-PEANUTS-compatibility tools.

Concretely, this three-tier architecture is implemented as a set of independent Python packages, one per node of figure 2, summarised in table 1 together with the responsibility of each.

Table 1: Packages composing `tpeanuts`, and their responsibility.

Package	Responsibility
<code>config/</code>	Composed configuration objects and named SM/BSM/NSI presets, replacing per-call keyword arguments.
<code>external/</code>	Optional third-party integrations (MCEq, nuSQuIDS, Honda, PyMSIS) for cross-validation and flux/density inputs.
<code>util/</code>	Device/dtype management (<code>RuntimeContext</code>) and generic numerical primitives.
<code>core/</code>	Medium-independent Hamiltonians, PMNS/mass-spectrum construction, evolution operators; standard and BSM ($3 + 1$ sterile, NSI, Majorana) variants.
<code>source/</code>	Emission models at the neutrino source: solar production profiles, reactor (Huber–Mueller) antineutrino spectra.
<code>medium/</code>	Environment-specific density/geometry models: vacuum, solar, Earth, atmosphere.
<code>detector/</code>	Forward-folding of oscillated flux into predicted event counts: cross sections, target composition, energy response, efficiency, backgrounds, per experiment.
<code>pipeline/</code>	Composition of the above into complete observables (solar/Earth/atmosphere detection chains).
<code>inference/</code>	Likelihood construction, LBFGS gradient-based fitting, Laplace/Fisher uncertainty estimation, likelihood scans.

4.1.1 Core, Domain and Application layers

Core Layer. At the base of the hierarchy, the core layer implements the medium-independent building blocks shared by every propagation scenario, for the plain three-flavour Standard Model and for the optional BSM extensions ($3 + 1$ sterile, NSI), together with an optional Majorana mass-matrix parametrisation relevant to lepton-number-violating observables rather than standard oscillation probabilities: the vacuum and matter Hamiltonians, PMNS/mass-spectrum construction, the matter potential, and the machinery to convert a Hamiltonian into a unitary evolution operator.

Two complementary propagation strategies are implemented here:

- **perturbative/analytical** evolver for smooth, slowly varying density profiles.
- **numerical** evolver, medium-independent, which discretises the trajectory, exponentiates a local Hamiltonian on each segment, and composes the segment operators through a logarithmic-depth binary-tree reduction, keeping the composition fully batchable and GPU-friendly.

Domain Layer. The domain layer specialises this shared core for each physical scenario, adding the neutrino-source, propagation-medium and detector models that the core layer itself knows nothing about. A source model isolates the physics of neutrino *emission*, upstream of propagation: solar production distributions from the Standard Solar Model, atmospheric neutrino profile production from MCEq/Honda, and the per-isotope Huber–Mueller (Huber, 2011; Mueller et al., 2011) reactor-antineutrino emission spectrum. A propagation-medium model then specialises the core with an environment-specific density/geometry treatment: vacuum, solar (adiabatic MSW/Landau–Zener mixing with production-point-averaged fluxes), Earth (matter regeneration along nadir-angle trajectories, with exposure time-averaged over the detector’s latitude and the Earth’s rotation), and atmosphere (propagation from a production height through the atmosphere).

Finally, a detector model folds an oscillated flux into a predicted, binned event count observable, entirely decoupled from how the probability was computed: a target composition, an energy response, an efficiency and a background, common to every experiment, combined with the relevant cross section by physical process: inverse beta decay to order $1/M$ (Vogel & Beacom, 1999) (reactor antineutrinos), neutrino–electron elastic scattering (solar), and deuteron dissociation (SNO). Five experiment-specific modules specialise this for Borexino, Daya Bay, IceCube DeepCore, and SNO phases I and II, summarised in table 2.

Table 2: Experiments implemented on the detector/inference layers, and their validation status against real published data (section 5.5.2).

Experiment	Channel / detector physics	Status against real data
Daya Bay	Reactor $\bar{\nu}_e$ IBD, 8 detectors, 6 reactors, full flux/response chain	Near/far fit recovers $\sin^2 2\theta_{13}$, Δm_{31}^2 jointly within 1–2% of published (section 5.5.1); full-shape fit not yet robust
IceCube DeepCore	Atmospheric ν_μ disappearance, event-by-event weighted-MC reweighting	Fit recovers θ_{23} , Δm_{3l}^2 within -0.05σ of published; fitted muon-background normalisation traced to fixed detector systematics
SNO II (salt phase)	CC/NC/ES, 38 published observables, published statistical covariance plus a reconstructed systematic covariance	Fit converges to θ_{12} , Δm_{21}^2 , λ_{8B} consistent with SNO’s own SNO-only range
Borexino	Solar ν – e elastic scattering, pp-chain spectrum	Direct fit of θ_{12} (and jointly with ε_{ee} , recovering the LMA-Dark degeneracy) to Borexino’s own published $P_{ee}(E)$ points succeeds; the full event-rate chain is implemented but its background model is still a placeholder, so no event-rate fit is yet reported
SNO (phase I)	CC/NC/ES, pure-D ₂ O phase	Day/night spectral fit to real published data recovers θ_{12} , within 0.86σ of NuFIT 6.1 data

Application Layer. At the top of the hierarchy, the application layer composes the models of the domain layer into complete, physically meaningful results. On one side, propagation pipelines assemble the source, medium and detector models into end-to-end observables, a solar-production-to-Earth-detection chain, or a full atmospheric flux propagated through the Earth.

On the other side, built on the same validated primitives, the statistical-inference machinery implements the formalism of section 2.4 as reusable, detector-agnostic components: a common differentiable-model interface, a free-parameter tuple and a prediction function, that every

concrete model, oscillation-only or detector-composed, implements, together with:

- **Likelihood functions**, a set of functions covering Poisson counts, asymmetric Gaussian ratios and, for covariance-matrix observables such as SNO II’s 38-observable data set, a correlated Gaussian likelihood, together with a Gaussian penalty term for externally constrained nuisance parameters;
- **LBFGS minimiser**, paired with the Laplace/Fisher covariance estimate;
- **2-dimensional likelihood-scan** routine, with an optional per-point re-minimisation of the remaining free parameters.

The concrete, experiment-specific models composing an oscillation model with a detector’s event-rate function live alongside each detector.

4.1.2 External-tool integrations

This layer isolates optional third-party dependencies from the core physics code:

- `external/mceq/` wraps the MCEq cascade-equation solver for atmospheric neutrinos (Fedynitch et al., 2015);
- `external/nusquids/` wraps nuSQuIDS (Arguelles et al., 2022) as an independent reference for validation and benchmarking;
- `external/honda/` reads the Honda/HKKM atmospheric-flux tables (Honda et al., 2007);
- `external/pymis/` provides an empirically calibrated atmospheric density model (Emmert et al., 2021).

Each is imported lazily so the package degrades gracefully when the optional extras are not installed.

4.1.3 Notebooks: validation, benchmarking, analysis, reproduction and inference

On top of the architecture described above, a curated suite of notebooks exercises every layer of `tpeanuts` and constitutes the primary, reproducible evidence base behind every result reported in this thesis. The suite is organised into five complementary groups.

- **Validation and benchmark** notebooks establish agreement against legacy PEANUTS and cross-validate against nuSQuIDS for every propagation medium, and measure CPU/GPU performance against both references.
- **Solar and atmospheric analysis** notebooks explore the fundamentals of each propagation regime, production and exposure profiles, differential-flux structure, and the behaviour of the underlying density models, independently of any specific detector or experiment.
- **Experiment Reproduction** notebooks recover well-known physical signatures, the solar day–night asymmetry, the MSW resonance, atmospheric oscillograms, Earth matter tomography, and the LMA-Dark degeneracy, establishing that the implementation reproduces established physics independently of any single reference code.
- **BSM extensions** notebooks validate the non-standard-interactions and sterile-neutrino extensions independently of the Standard-Model reproduction above: NSI notebooks compare the solar ^8B survival probability under literature NSI presets against the Standard-Model limit and scan individual off-diagonal couplings; sterile-neutrino notebooks confirm

the $3 + 1$ extension's Standard-Model limit, unitarity and vanishing sterile matter coupling under the CC-only default, together with its expected CP-violation and short-baseline sensitivity.

- **Inference** notebooks close the chain, running real-data prediction checks, normalisation audits and parameter-recovery fits where the experimental model is sufficiently complete (Daya Bay, IceCube DeepCore, SNO phase I and SNO phase II); for Borexino, a direct oscillation-parameter fit to the published $P_{ee}(E)$ points succeeds, while the full detector-level event-rate chain is validated only at component and unit-test level.

4.2 Numerical simulation conventions

The numerical core of **tpeanuts** is built around a small number of deliberate conventions that keep every propagation medium numerically consistent, dimensionally well-behaved, and directly comparable to both the legacy PEANUTS implementation and to **nuSQuIDS**. The most important of these are the choice of a unitary, dimensionless evolution scale for the Hamiltonian, the neutrino/antineutrino sign convention, and the floating-point precision policy used throughout the matrix-exponential evolution.

4.2.1 Non-dimensionalisation of the Hamiltonian

Rather than working directly with the physical matter potential and kinetic (mass-squared) terms, which span many orders of magnitude depending on the neutrino energy and the medium's density, every Hamiltonian contribution is first converted into a dimensionless quantity by multiplying it by a configurable evolution length scale, `evolution_scale_m` (the Earth radius, by default). Concretely,

$$V = s L_{\text{scale}} \sqrt{2} G_F N_A (10^6 \text{ cm}^3/\text{m}^3) (\hbar c)^2 n_e, \quad k_i = L_{\text{scale}} \frac{\Delta m_i^2}{2E\hbar c}, \quad (21)$$

where $s = \pm 1$ is the neutrino/antineutrino sign (see below), N_A is Avogadro's number and the $10^6 \text{ cm}^3/\text{m}^3$ factor converts n_e , the electron number density in mol/cm^3 , to mol/m^3 consistently with L_{scale} in metres, and Δm_i^2 are the (reduced) mass-squared eigenvalues. The trajectory coordinate is normalised by the same scale, $x = L/L_{\text{scale}}$, so that the evolution operator is always computed as $U = \exp(-iHx)$, with both H and x dimensionless.

This convention avoids mixed physical length units across media without changing the physical phase $\hat{H}x = H_{\text{phys}}L$, since only the ratio L/L_{scale} enters the exponent; it defaults to the Earth's radius for backward compatibility with legacy PEANUTS, but is an explicit, per-medium parameter, settable e.g. to the injection height of an atmospheric shower rather than to $R_{\oplus}$.

4.2.2 Neutrino/antineutrino convention

Antineutrino propagation is selected through a single boolean (or boolean-tensor, for batched mixed samples) flag, `antinu`, carried by the `OscillationParameters` object and threaded through every Hamiltonian and probability routine. Two independent sign flips are applied consistently whenever `antinu=True`:

- **Matter potential.** The Wolfenstein charged-current potential changes sign, $V \rightarrow -V$, since it originates from the coupling of ν_e (not $\bar{\nu}_e$) to the ambient electrons.

- **Mixing matrix.** The PMNS matrix is replaced by its complex conjugate, $U \rightarrow U^*$, equivalent to flipping the sign of the CP-violating phase, $\delta_{\text{CP}} \rightarrow -\delta_{\text{CP}}$.

Both substitutions are applied at the point where the reduced mixing matrix and the matter Hamiltonian are assembled, so that every downstream propagation routine, vacuum, solar, Earth or atmospheric, inherits the correct antineutrino behaviour automatically from a single flag, without medium-specific antineutrino logic. The same object also encodes the neutrino mass ordering as the sign of Δm_{3l}^2 : positive for normal, negative for inverted ordering.

4.2.3 Precision and evaluation mode

All propagation is carried out at double precision (`torch.float64` real / `torch.complex128` complex) by default, matching the precision of the legacy NumPy implementation and keeping the matrix-exponential composition numerically stable over trajectories segmented into hundreds of steps. The fast-forward simulation pipelines are wrapped in `@torch.no_grad()` by default, since most forward evaluation, benchmarking and physics reproduction does not need gradients and disabling autograd bookkeeping reduces memory footprint and computational overhead; the underlying physics primitives themselves, and the experiment-specific inference models built directly on them (section 4.1.1), do not disable autograd, and it is this differentiable path, not the fast pipelines, that drives the gradient-based fitting of section 3.4.

For validation purposes, a `legacy_precision` flag is threaded through the potential and Hamiltonian builders; when enabled, it substitutes the full-precision matter-potential prefactor, derived from CODATA G_F , N_A and $\hbar c$ (Tiesinga et al., 2021), with the four-significant-digit constant hard-coded in the original PEANUTS implementation, allowing bit-level, agreement to be verified against the legacy reference where required.

4.3 Neutrino detection pipelines

Building on the medium-specific propagation primitives described above, the framework assembles them into two complete, physically distinct detection workflows: a solar-neutrino pipeline, which propagates neutrinos produced deep inside the Sun to a terrestrial detector, and an atmospheric-neutrino pipeline, which propagates neutrinos produced by cosmic-ray air showers in the Earth’s atmosphere down to a detector. Both share the same underlying core Hamiltonian/evolutor machinery; they differ in whether the mass-eigenstate coherence acquired at production survives all the way to the detector.

4.3.1 Solar-neutrino detection pipeline

A solar neutrino is produced as a pure ν_e at some radius ρ inside the Sun, distributed according to a source-specific production profile (e.g. ${}^8\text{B}$, pp) (Vinyoles et al., 2017). As it propagates outward through the Sun’s smoothly varying electron density, the MSW matter effect adiabatically rotates the flavour state into the local mass-eigenstate basis, with a small non-adiabatic level-crossing correction (Parke, 1986). Because the $\sim 1.5 \times 10^8$ km vacuum baseline to Earth is enormously larger than the oscillation length at solar-neutrino energies, the relative phases between mass eigenstates wash out completely before arrival: only the incoherent mass-eigenstate populations survive.

`tpeanuts` implements this picture as a pipeline reproducing, in batched `torch` form, the

original PEANUTS NumPy/Numba reference workflow used to validate it bit-for-bit. The pipeline proceeds in three stages:

1. **Solar production and adiabatic conversion:** for a given source and energy grid, the electron-neutrino state is propagated adiabatically outward from each production radius ρ in the source's normalised production-radius distribution, and the result is averaged over that distribution to yield a single vector of mass-eigenstate weights, $w_i(E) = (w_1, w_2, w_3)$, the fraction of the flux exiting the Sun in each mass eigenstate.
2. **Vacuum propagation to Earth:** because no relative phase is ever tracked, propagation across the Sun–Earth vacuum baseline leaves the mass weights $w_i(E)$ unchanged by construction; this step leaves the mass weights exactly unchanged, an exact identity of the incoherent formulation with no separate approximation involved.
3. **Earth matter propagation:** at the detector, the incoherent mixture of incident mass eigenstates is propagated through Earth matter, relevant only for underground detectors sensitive to the day/night matter-regeneration asymmetry, evaluated over a grid of nadir angles η ; each mass eigenstate acquires its own Earth-matter transition probability to each final flavour α , $P_{i \rightarrow \alpha}^\oplus(E, \eta)$, and the final flavour probability is their incoherent sum, $P_{\nu_e \rightarrow \nu_\alpha}(E, \eta) = \sum_i w_i(E) P_{i \rightarrow \alpha}^\oplus(E, \eta)$.

This oscillation probability, still a function of energy rather than a detector observable, is further averaged: over the solar production-radius distribution (folded into the first stage), over the neutrino energy spectrum, and, for underground detectors, over the detector's nadir-angle exposure, the fraction of a year during which each nadir angle is observed, computed from the detector's geographic latitude and the Earth's daily and annual motion; converting it into a predicted event rate additionally requires the flux, cross section and detector response.

4.3.2 Atmospheric-neutrino detection pipeline

Atmospheric neutrinos follow a different physical picture. They are produced across a broad, energy- and angle-dependent distribution of production heights h in the atmosphere, instead of at a single well-defined point, with the boundary-condition flux $\Phi_\beta(E, h, \theta)$ for each produced flavour β supplied by an external cascade-equation generator (MCEq Fedynitch et al., 2015) or by tabulated Honda/HKKM fluxes (Honda et al., 2007), which provide ν_e , $\bar{\nu}_e$, ν_μ and $\bar{\nu}_\mu$; the negligible conventional ν_τ flux is carried with zero initial value. Unlike the solar case, coherence is not lost here: even for an upward-going, Earth-crossing trajectory of up to $\sim 1.3 \times 10^4$ km (section 2.1), the wave-packet coherence length at GeV-scale atmospheric-neutrino energies vastly exceeds this baseline (Giunti & Kim, 2007), so the mass-eigenstate wave packets remain overlapped and the full flavour-amplitude state is propagated coherently from production to detector. The pipeline proceeds in five stages:

1. **Flux selection:** one produced particle and one detector zenith angle are sliced out of the loaded height/energy/angle flux dataset.
2. **Atmosphere propagation:** a coherent unit-amplitude flavour state $|\beta\rangle$ is evolved from each production height down to the Earth's surface, batched jointly over the full energy $\times$ height grid.

3. **Earth propagation:** for an upward-going, Earth-crossing trajectory, the surface state is propagated further through Earth matter to the detector depth, using the same coherent evolver as the solar pipeline’s coherent variant.
4. **Height integration:** the detector-level flux for each final flavour α is obtained by weighting the propagated $\beta \rightarrow \alpha$ probability at each height by $\Phi_\beta(E, h, \theta)$ and trapezoidally integrating over h .
5. **Flavour summation:** the height-integrated contributions from every produced flavour are summed into the total detector-level flux of each final flavour.

4.4 Test suite and sanity checks

The automated `tpeanuts` test suite contains 942 collected test cases in 64 files organised according to the package module tree. It is run manually with `pytest` after relevant modifications. These tests overlap across three broad categories, supplemented by a further, notebook-level layer of validation that `pytest` does not cover:

- **Code consistency checks** cover output shape, dtype and device, broadcasting over batched grids, invalid inputs, alternative analytical and numerical paths, and save/load round trips. They account for roughly a third of the suite.
- **Mathematical checks** compare outputs with independently derived expressions or algebraic identities. Examples include the PMNS determinant and Jarlskog invariant, manually accumulated products of segment evolvers, constant-density comparisons of the perturbative and numerical methods, the closed-form resonance radius, and the scaling of non-dimensionalised terms. These checks represent approximately a quarter of the suite.
- **Physical coherence checks** test probability conservation, unitarity, known physical limits such as the adiabatic solar limit, and individual functions against the original PEANUTS implementation. Because categories overlap, these tests also comprise roughly a third to two-fifths of the suite.
- **Notebook-level validation** compares legacy PEANUTS at module-dependent precision and `nuSQuIDS` at the scenario-dependent tolerances of section 3.1.2. The notebooks also contain the real-data prediction checks, normalisation audits and parameter-recovery fits, together with inline smoke-test assertions.

Detector and inference validation concentrates on code-consistency and mathematical properties because no independent implementation is available for direct comparison. Closure tests and parameter-recovery fits are instead run in the notebooks described in section 3.4.

5 Validation and Results

This section presents the validation, benchmark, simulation and phenomenological results obtained for `tpeanuts`, following the methodology of section 3.

Section 5.1 establishes numerical reliability against PEANUTS and `nuSQuIDS` and computational usefulness via CPU/GPU benchmarks. Section 5.2 then exercises capabilities beyond a PEANUTS translation, vacuum, solar and atmospheric-neutrino propagation through the

Earth with flexible matter models, before section 5.3 reproduces well-known physical phenomena, section 5.4 extends this to non-standard interactions and sterile neutrinos, and section 5.5 infers oscillation parameters directly from real, published experimental data. Section 5.6 documents the public GitHub code release underlying all of it.

5.1 Validation and benchmark

5.1.1 Internal validation and benchmark

The perturbative/analytical and the purely numerical evolvers of section 2.1 solve the same propagation problem through independent numerical strategies and must therefore agree with each other. The same self-consistency logic also applies to the discrete implementation choices within each evolver, reunitarizing the truncated perturbative operator or not, the closed-form (Cardano) versus dense eigenvalue solver, the Landau–Zener correction, and the runtime context (float32/float64 precision, CPU/CUDA device). Table 3 summarises these checks for the Standard Model.

Table 3: Standard Model intrinsic self-consistency: maximum and mean absolute and relative errors.

Comparison	max abs	mean abs	max rel	mean rel
Solar: <code>adiabatic_approximated</code> vs <code>numerical</code>	4.88×10^{-3}	3.85×10^{-4}	4.58×10^{-2}	2.50×10^{-3}
Solar: <code>adiabatic_exact</code> vs <code>numerical</code>	4.87×10^{-3}	3.80×10^{-4}	4.56×10^{-2}	2.47×10^{-3}
Earth: perturbative vs numerical	3.22×10^{-3}	1.32×10^{-4}	9.24×10^{-3}	3.96×10^{-4}
Atmosphere: perturbative vs numerical	1.44×10^{-11}	1.86×10^{-13}	1.07×10^{-8}	1.76×10^{-11}
Reunitarize off vs on	1.50×10^{-2}	1.01×10^{-3}	2.05×10^{-2}	3.26×10^{-3}
Eigenv.: Cardano vs <code>torch.linalg.eigvalsh</code>	9.06×10^{-13}	1.56×10^{-14}	1.63×10^{-11}	1.19×10^{-13}
Landau–Zener on vs off (solar surface)	0	0	0	0
Precision: numerical float32 vs float64	1.04×10^{-4}	1.11×10^{-5}	1.75×10^{-4}	3.35×10^{-5}
Device: numerical CPU vs CUDA (float64)	4.92×10^{-14}	2.38×10^{-15}	3.06×10^{-13}	1.02×10^{-14}

For the NuFIT 5.2 preset, both adiabatic solar methods differ from the numerical calculation by absolute errors of $\mathcal{O}(10^{-3})$ and the Landau–Zener correction produces no appreciable change in this test. The perturbative Earth evolver differs from the numerical one at the same 10^{-3} level, with smaller errors after reunitarization, whereas the atmospheric comparison agrees to 10^{-8} or better. Cardano/Ferrari eigenvalues agree with `torch`’s diagonalization close to floating-point precision. Moving the float64 calculation between CPU and CUDA introduces differences below 10^{-12} , while float32 discrepancies reach 10^{-4} . On the grids tested here, the combined fast path—adiabatic propagation in the Sun and reunitarized perturbative propagation through the Earth or atmosphere—has inaccuracies dominated by the $\mathcal{O}(10^{-3})$ approximation error compared with the numerical path.

Figure 3 complements this precision picture with the matching wall-clock cost, for the same four representative workloads: pure-flavour Earth propagation, solar production to detector, and atmosphere propagation to the surface and to the detector. At matched grid sizes the perturbative evolver is consistently faster than the purely numerical one, with a median speed-up of $2.8\text{--}4.0\times$ across the four workloads and a maximum of $5\text{--}6\times$ (up to $6.1\times$ for the solar pipeline) at the largest grid tested (64×64), for the same 10^{-3} -level agreement already quantified in table 3.

Table 4 isolates the two runtime properties, device and precision, on the fixed solar-to-Earth-detector workload used throughout, holding everything else at its default, and repeats the method on that same fixed workload as a cross-check. Two results stand out. First, moving this workload

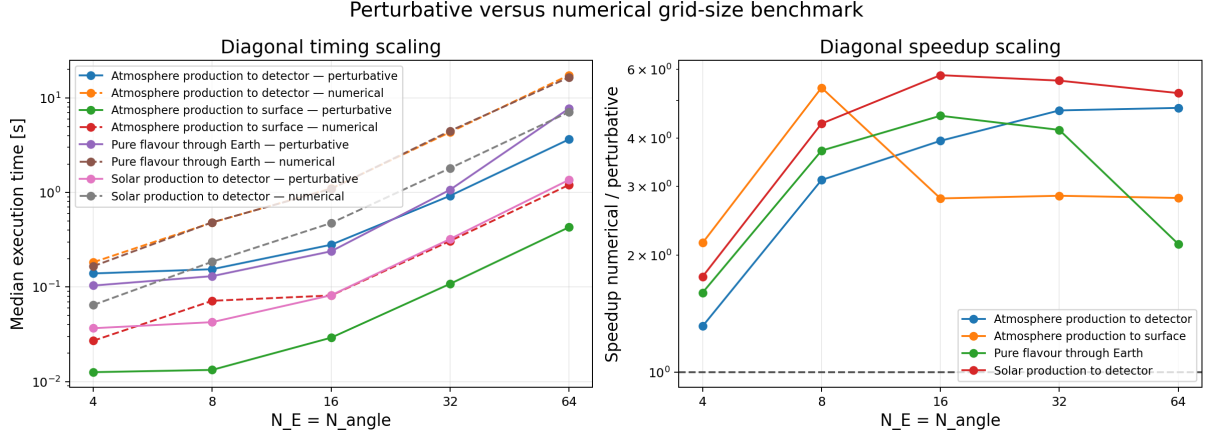


Figure 3: Perturbative vs. numerical evolver: median wall-clock time (left) and speed-up (right) vs. grid size, for four representative propagation workloads.

from CPU to CUDA is not a speed-up at either grid size tested here: the GPU runs at $0.44\text{--}0.50\times$ the CPU wall-clock time, i.e. $2\text{--}2.3\times$ *slower*, confirming the kernel-dispatch-overhead argument of section 3.1.3, a 64×64 grid is still too small to amortise it for a computation this cheap. Second, float32 is essentially neutral at the smallest grid ($0.99\times$) and only pays off, modestly, at the largest one ($1.33\times$), for the same reason.

Table 4: Intrinsic runtime-axis benchmark on the fixed solar-to-Earth-detector workload: wall-clock time and speed-up at the smallest and largest grid tested.

Axis	Grid	Baseline [s]	Candidate [s]	Speed-up
Device: CPU vs. CUDA (<code>float64</code>)	4×4	0.0177 (CPU)	0.0400 (CUDA)	$0.44\times$
	64×64	0.794 (CPU)	1.581 (CUDA)	$0.50\times$
Precision: float64 vs. float32	4×4	0.0447 (float64)	0.0452 (float32)	$0.99\times$
	64×64	1.717 (float64)	1.290 (float32)	$1.33\times$
Method: numerical vs. fast path	4×4	0.2506 (numerical)	0.0374 (fast)	$6.70\times$
	64×64	7.471 (numerical)	1.302 (fast)	$5.74\times$

5.1.2 PEANUTS validation and benchmark

Bit-level agreement with the legacy PEANUTS implementation is the correctness target for the translation, and is reached only for the `solar` module, agreeing to $\sim 5.5 \times 10^{-15}$ relative, at machine precision. Figure 4 (left) summarises the maximum absolute and relative differences observed across the four validated modules, solar, Earth, vacuum and the composed detection pipeline, when both codes are evaluated with the `legacy_precision` flag enabled on matched B16-AGSS09 input tables for the solar-dependent modules. All four modules agree to $\sim 10^{-7}$ or better in *relative* terms (Earth $\sim 5.4 \times 10^{-10}$, the composed pipeline $\sim 2.6 \times 10^{-9}$, vacuum $\sim 1.5 \times 10^{-7}$), comfortably at or below the 1 ppm reference line on that metric; the corresponding *absolute* differences for the pipeline and vacuum modules are larger, $\mathcal{O}(10^{-4})$ and $\mathcal{O}(10^{-5})$ respectively, above the 1 ppm line shown in the figure, reflecting the larger physical scale of those observables rather than a failure of relative agreement.

All wall-clock benchmarks reported in this thesis, against both legacy PEANUTS and `nuSQuIDS`, were run on a single workstation: an AMD Ryzen 7 1800X eight-core/16-thread CPU, 32 GB of RAM, and an NVIDIA GeForce GTX 1080 Ti GPU (11 GB VRAM) for the CUDA comparisons. The reported speed-ups therefore characterise this specific, comparatively modest and by-now dated hardware rather than a hardware-independent property of the im-

plementations; a current CPU or a more recent GPU would be expected to shift the absolute numbers while leaving the qualitative scaling trends unchanged.

Figure 4 (right) gives the wall-clock speed-up of **tpeanuts** over legacy PEANUTS for Earth propagation as the energy and nadir grids increase. At 4×4 , **tpeanuts** is up to ~ 10 times slower. It becomes faster once the grid contains a few tens of points per dimension, peaks at $\sim 5.7\times$ near 512×512 , and falls slightly to $\sim 4.9\times$ at 1024×1024 . PEANUTS already vectorises NumPy operations over a trajectory; the tensor implementation gains an advantage only when the batch is large enough to amortise the fixed PyTorch dispatch overhead described in section 3.1.3.

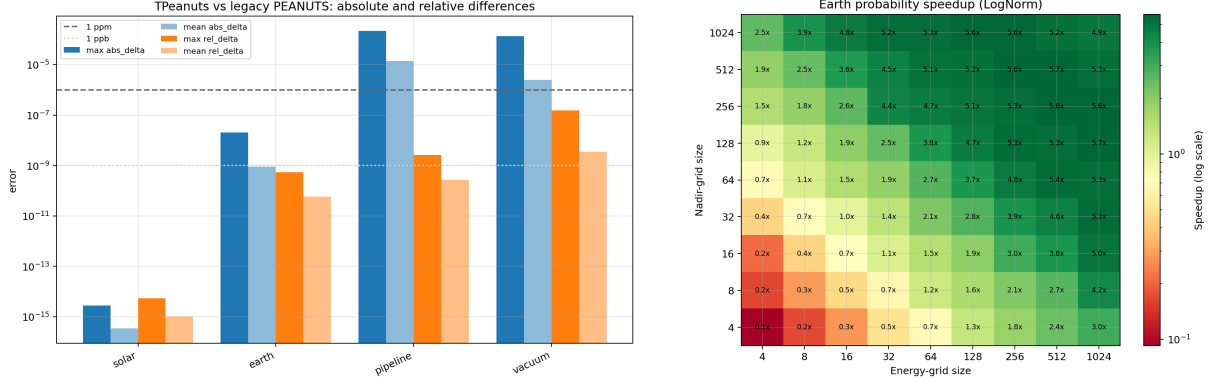


Figure 4: PEANUTS validation: absolute/relative differences by module, 1 ppm/1 ppb reference lines (left); wall-clock speed-up over PEANUTS for Earth propagation vs. grid size (right).

5.1.3 nuSQuIDS validation and benchmark

Figure 5 (left) summarises the maximum absolute and relative differences between **tpeanuts** and **nuSQuIDS** across four different configurations: vacuum, atmosphere, solar and the composed solar-to-Earth-detection pipeline (**SolarNeutrino**). The vacuum and atmosphere modules agree to within 10^{-8} – 10^{-7} in relative terms, comfortably below the 1 ppm reference line, as expected for two independent, essentially exact solvers of the same closed propagation problem. The solar module and the composed solar-Earth-detection pipeline instead show markedly larger discrepancies, at the percent ($\sim 10^{-2}$) level, several orders of magnitude above the vacuum/atmosphere agreement.

Unlike the vacuum and atmospheric cases, the solar and solar–Earth observables depend on production-averaged incoherent mass-eigenstate weights and on the treatment of coherence loss between the Sun and the Earth. **tpeanuts** follows the corresponding PEANUTS conventions (section 2.1), whereas the independent **nuSQuIDS** calculation need not make identical choices. These differences are plausible sources of the observed disagreement, although their individual contributions are not isolated here through controlled ablations. The comparison nevertheless reveals a discrepancy that is absent when **tpeanuts** is compared with PEANUTS, where these conventions are shared.

Figure 5 (right) shows the wall-clock speed-up of **tpeanuts** over **nuSQuIDS** for the integrated solar-detector probability calculation, as a function of the energy-grid and nadir/ $\cos \theta_z$ -grid sizes used in the batched evaluation. The speed-up grows monotonically with grid size, from $\sim 63\times$ for the smallest grid considered (2×2) to over $1100\times$ for the largest (32×32), directly confirming the scaling behaviour anticipated in section 3.1.3: because **nuSQuIDS** evaluates one solver object per energy/angle point while **tpeanuts** evaluates the full grid in a single batched call, the relative

advantage of the tensor-based formulation increases with the number of configurations evaluated per call.

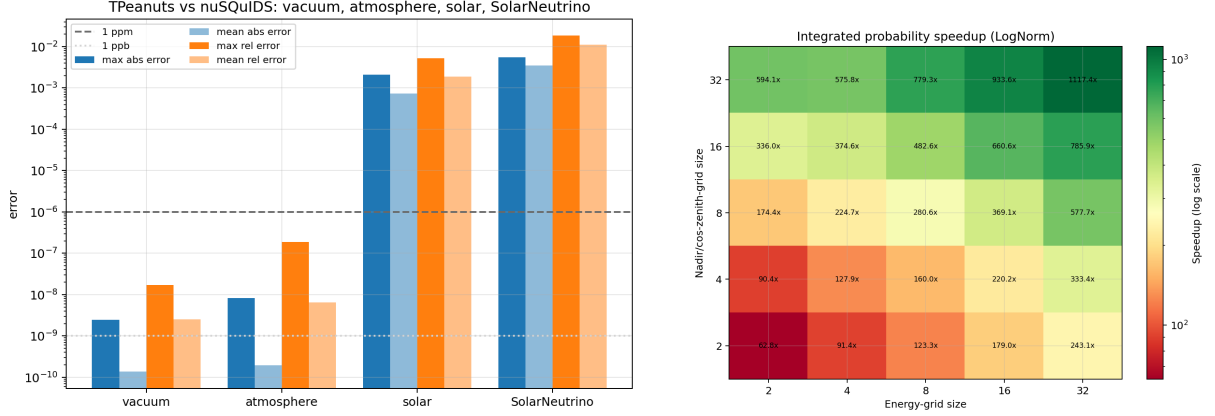


Figure 5: nuSQuIDS validation: absolute/relative differences by configuration, 1 ppm/1 ppb reference lines (left); wall-clock speed-up over nuSQuIDS for the integrated solar-detector probability vs. grid size (right).

5.2 Simulation analysis

This subsection applies the propagation engine to vacuum, solar- and atmospheric-neutrino calculations. The examples cover complete chains, from production to Earth regeneration where applicable, rather than isolated probabilities. Their outputs are compared with qualitative or quantitative expectations from standard oscillation phenomenology. The specific experimental signatures considered with the same code are reported in section 5.3.

5.2.1 Vacuum neutrino

Figure 6 maps the vacuum survival probability $P(\nu_e \rightarrow \nu_e)$ and appearance probability $P(\nu_\mu \rightarrow \nu_e)$ over $L = 1\text{--}1.5 \times 10^4$ km and $E = 0.1\text{--}10$ GeV. This range includes reactor, accelerator and atmospheric configurations. Both channels contain the oscillatory phase $1.267 \Delta m_{31}^2 L/E$, and their diagonal fringes consequently follow similar L/E patterns. Unitarity relates $1 - P_{ee}$ to $P_{e\mu} + P_{e\tau}$, and only to $P_{\mu e}$ in the two-flavour limit, so the two channels are closely related without being exactly complementary. In the left panel, P_{ee} is close to unity in the light-blue regions and decreases towards its first oscillation minimum in the dark-blue bands. The corresponding regions in the right panel show small and large $P_{\mu e}$, respectively. This behaviour is consistent with two-flavour-dominated conversion at fixed L/E when the sub-leading solar splitting is neglected.

The chequerboard structure at the largest L/E is a beat pattern between the atmospheric and solar mass splittings. In this region the solar phase $\Delta m_{21}^2 L/E$ completes many cycles, and the combined pattern aliases against the finite plotting grid. Since every pixel uses the full three-flavour vacuum evolution operator, the map checks the L/E scaling and the relative survival and appearance amplitudes. The CP-violation calculation in figure 14 separately tests the phase dependence of the same operator.

5.2.2 Solar neutrino

Figure 7 shows the Earth-regeneration probability $P(\nu_\odot \rightarrow \nu_\alpha; E)$ for incoherent solar ${}^8\text{B}$ mass-basis weights propagated along four nadir-angle trajectories. These cross the inner core, outer core, mantle or crust only and are compared with the no-Earth vacuum reference. Below ~ 4 MeV,

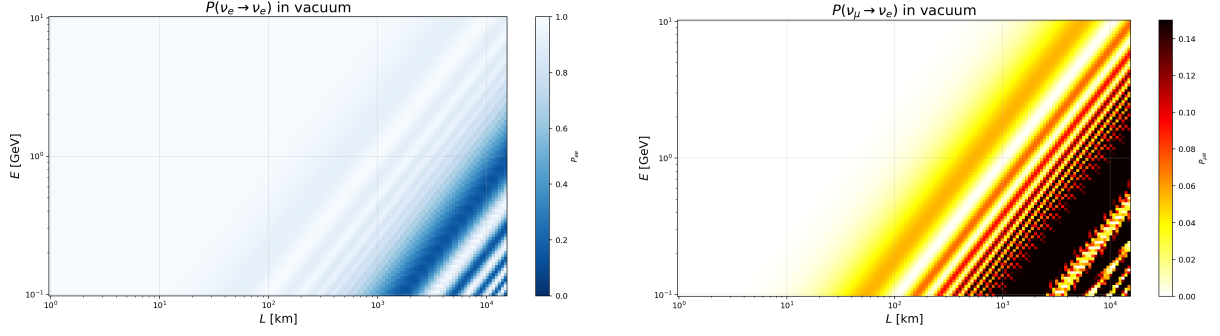


Figure 6: Vacuum oscillation probabilities over the (L, E) plane: survival $P(\nu_e \rightarrow \nu_e)$ (left) and appearance $P(\nu_\mu \rightarrow \nu_e)$ (right).

all trajectories nearly overlap the vacuum curve. In this range, the ratio $2EV_{CC}/\Delta m^2$, which controls day–night regeneration, remains small compared with unity. Above ~ 6 MeV, the curves separate from the vacuum reference and from one another. The largest difference occurs for the inner-core trajectory, which also has the longest and densest matter path. The same effect is quantified through the day–night asymmetry in section 5.3.1. Numerically, the three channels satisfy $\sum_\alpha P(\nu_\odot \rightarrow \nu_\alpha) = 1$ at every sampled energy and trajectory.

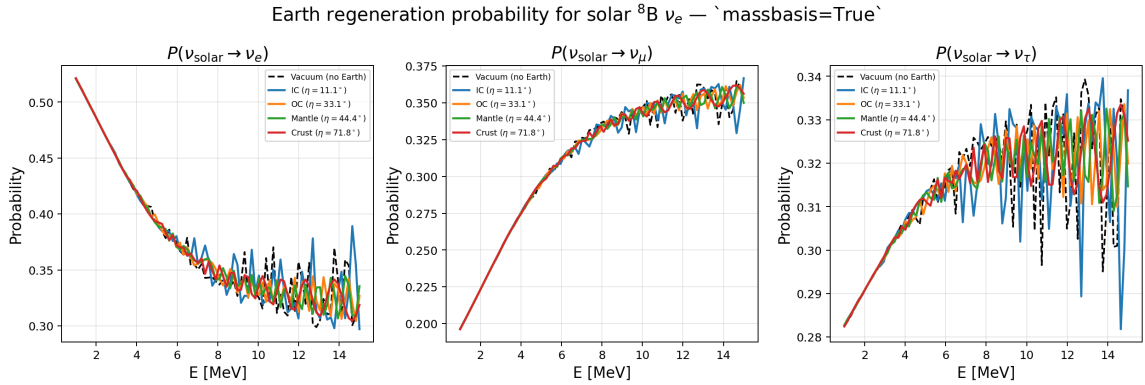


Figure 7: Earth-regeneration probability for solar ${}^8\text{B}$ neutrinos produced as incoherent mass-basis weights, $P(\nu_\odot \rightarrow \nu_\alpha; E)$, for four representative nadir-angle trajectories crossing the inner core, outer core, mantle and crust, compared with the no-Earth vacuum reference.

Figure 8 combines the continuous pp , hep , ${}^8\text{B}$, CNO channels and the monoenergetic ${}^7\text{Be}$ lines; pep is not plotted. Each continuous source is weighted by its normalised spectrum $g_s(E)$ and by the corresponding Earth-regeneration probability from figure 7. The resulting differential flux $d\Phi_\alpha^{\text{tot}}/dE(E, \eta_{\text{fix}})$ vanishes outside the tabulated range of each source. The ${}^7\text{Be}$ lines at 0.384 and 0.862 MeV are drawn separately; their heights represent integrated line fluxes rather than spectral densities. Below the pp endpoint, ~ 0.42 MeV, the spectrum is of order $10^{10} \text{ cm}^{-2} \text{ s}^{-1} \text{ MeV}^{-1}$. Above it, the hep , ${}^8\text{B}$ and CNO continuum is five to six orders of magnitude smaller, near $10^5 \text{ cm}^{-2} \text{ s}^{-1} \text{ MeV}^{-1}$ between 3 and 10 MeV, and falls towards $10 \text{ cm}^{-2} \text{ s}^{-1} \text{ MeV}^{-1}$ at 18 MeV. The ν_e component dominates at low energy, while the three flavour fluxes approach one another above ~ 9 MeV.

Isolating the ${}^8\text{B}$ channel and exercising the nadir-angle exposure integration for four representative surface latitudes gives three flavour spectra that are, to the resolution of the plot, indistinguishable across all four (figure 8, right): annual averaging over the day/night nadir-angle exposure smooths out the latitude-dependent matter effects seen in the left-hand spectrum, as

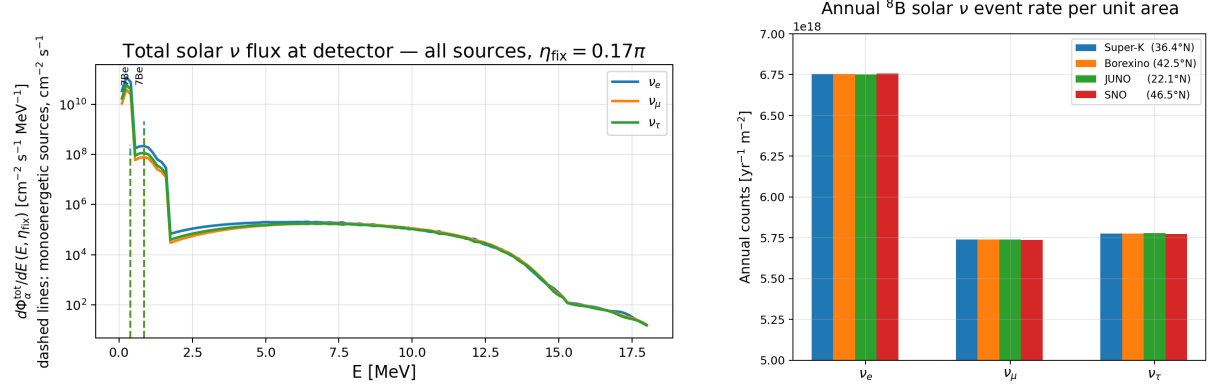


Figure 8: Total solar-neutrino differential flux reaching the detector on the Earth surface, $d\Phi_{\alpha}^{\text{tot}}/dE(E, \eta_{\text{fix}})$, summed over the continuous solar production channels weighted by their normalized production spectra (dashed vertical lines: the monoenergetic ${}^7\text{Be}$ transitions, shown at their own oscillated flux rather than a density), at fixed nadir angle $\eta_{\text{fix}} = 0.17\pi$ (left); annual, exposure-integrated ${}^8\text{B}$ event rate per unit area, by flavour, at four detector latitudes (right).

expected from the small latitude differences considered. This annual event rate corresponds to the full terrestrial propagation of solar neutrinos, from production through Earth matter to the detector, evaluated with the perturbative (analytical) evolver of section 2.1.

5.2.3 Atmospheric neutrino

The atmospheric detection pipeline (section 4.3) requires, as a boundary condition, the production-height distribution of the parent flux at each energy and angle. Figure 9 compares this distribution, $f(h|E, \alpha)$, for ν_{μ} at a fixed zenith angle ($\alpha = 31.8^\circ$) as generated by MCEq (Fedynitch et al., 2015) and by the tabulated Honda/HKKM parametrisation (Honda et al., 2007). Both place the bulk of the production between ~ 10 and 30 km, peaking around 15 km and shifting slightly higher at higher energy; the visibly discretised Honda distribution reflects its coarser production-height binning, but the two agree closely in shape and normalisation.

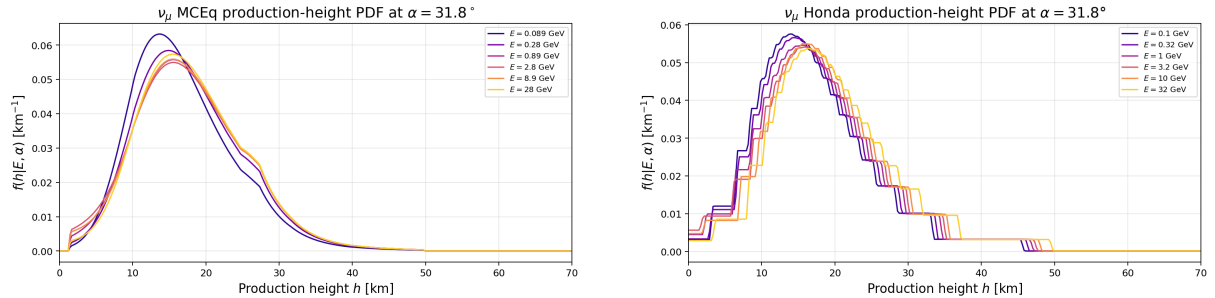


Figure 9: ν_{μ} production-height probability density, $f(h|E, \alpha)$, at fixed zenith angle $\alpha = 31.8^\circ$, generated with MCEq (left) (Fedynitch et al., 2015) and with the Honda/HKKM parametrisation (right) (Honda et al., 2007).

Figure 10 gives the surface probabilities $P(\bar{\nu}_{\mu} \rightarrow \bar{\nu}_{\alpha}; E, h)$ for an antimuon-neutrino primary at $\theta = 18.2^\circ$. They are evaluated as functions of energy and production height, before applying the height distribution of figure 9 or propagating through the Earth. Above ~ 0.3 GeV, survival dominates most of the plane, with $P(\bar{\nu}_{\mu} \rightarrow \bar{\nu}_{\mu}) \gtrsim 0.8$. The appearance channels are concentrated below ~ 0.3 GeV, in a band that widens with production height. This dependence follows from the longer atmospheric baseline available to neutrinos produced at greater height and is consistent with the expected L/E behaviour before Earth matter effects are included.

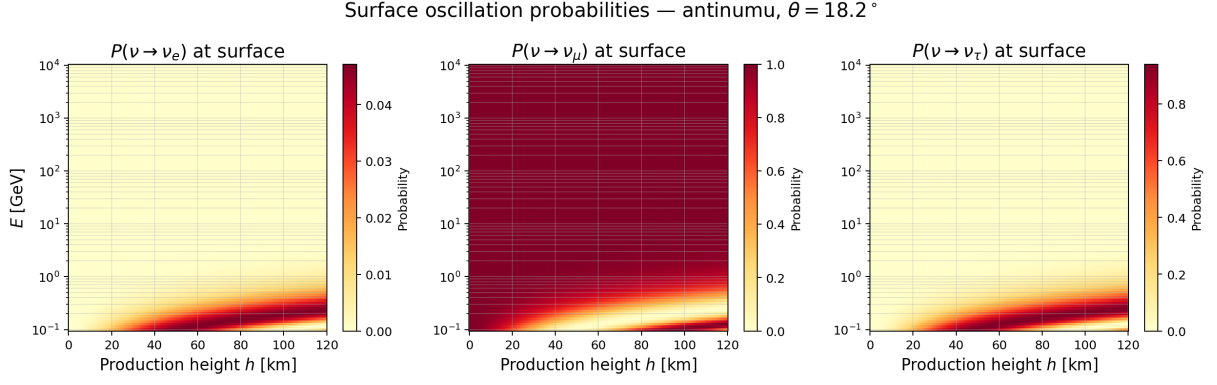


Figure 10: Surface oscillation probabilities $P(\bar{\nu}_\mu \rightarrow \bar{\nu}_\alpha; E, h)$ for an antineutrino primary at fixed zenith angle $\theta = 18.2^\circ$, as a function of energy and production height.

After propagation through the atmosphere and Earth, and averaging over the production-height distribution of figure 9, the detector-level survival probability $P_{\text{det}}(\nu_\mu \rightarrow \nu_\mu; E, \theta)$ is shown in figure 11. It remains close to unity above 100 GeV and decreases to approximately 0.1–0.2 at sub-GeV energies and large zenith angles. On this grid, the height-integrated and flavour-summed calculation is consistent with the expected atmospheric ν_μ disappearance pattern.

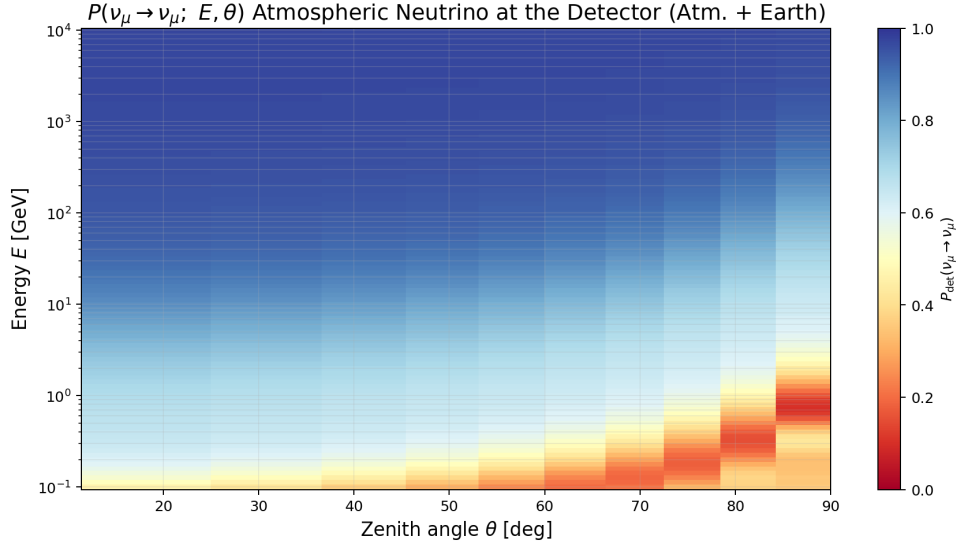


Figure 11: Height-averaged, detector-level ν_μ survival probability $P_{\text{det}}(\nu_\mu \rightarrow \nu_\mu; E, \theta)$, combining atmospheric and Earth propagation.

5.3 Reproduction of experimental results

This subsection examines six experimental or phenomenological signatures: the solar day–night asymmetry, the adiabatic approximation, the MSW resonance, CP violation, atmospheric oscillograms and Earth matter tomography. Each calculation is compared with a published measurement or an analytic expectation, as specified in the corresponding subsection.

5.3.1 Solar day–night asymmetry

Earth regeneration of solar ν_e during night-time observation is quantified through $A_{\text{DN}} = 2(P_{ee}^{\text{day}} - P_{ee}^{\text{night}})/(P_{ee}^{\text{day}} + P_{ee}^{\text{night}})$. For the ^8B flux and the NuFIT 5.2 normal-ordering parameters (Esteban et al., 2020), the flux-weighted, energy-integrated prediction at the Super-Kamiokande site is $A_{\text{DN}}^{\text{int}} = -2.69\%$. This value lies within the 1σ interval of the combined Super-Kamiokande

measurement, $(-3.2 \pm 1.1_{\text{stat}} \pm 0.5_{\text{syst}})\%$ (Renshaw et al., 2014). At the precision of this comparison, the pipeline prediction is therefore compatible with the observed regeneration signal.

A more detailed, detector-specific comparison additionally accounts for two effects relevant to the Super-Kamiokande measurement: its 3.5 MeV electron-recoil cut corresponds to a higher effective neutrino-energy threshold, $E_\nu^{\text{eff}} \approx 5.5$ MeV, and the observed rate is weighted by the full elastic-scattering cross section rather than by the neutrino spectrum alone. Table 5 shows that both corrections push the predicted asymmetry closer to the measured value, from -2.69% to -3.20% , in better agreement with the Super-Kamiokande measurement and consistent with the mild ($\sim 1.3\sigma$) tension already present between that measurement and the theoretical LMA-MSW range (computed directly from the standard LMA parameters, not an external published value).

Table 5: Day–night asymmetry $A_{\text{DN}}^{\text{int}}$ at Super-Kamiokande: as computed ($E_\nu > 3.5$ MeV threshold, spectral weight only) and with the two Super-Kamiokande-specific corrections applied, compared against the independent LMA-MSW prediction and the combined Super-Kamiokande measurement.

Detector	tpeanuts	tpeanuts-sk	LMA std.	Published
Super-K	-2.69%	-3.20%	-2.0% to -1.7%	$(-3.2 \pm 1.1_{\text{stat}} \pm 0.5_{\text{syst}})\%$

5.3.2 Adiabatic approximation

The adiabatic approximation is assessed with the parameter $\gamma(\rho)$, which must remain well above the non-adiabatic onset at $\gamma = 1$ for the adiabatic MSW treatment to apply (Haxton, 1986; Parke, 1986). Following Parke’s formalism, γ is evaluated at the MSW resonance layer as

$$\gamma_{\text{res}}(E) = \frac{\Delta m_{21}^2 \sin^2 2\theta_{12}}{2E \hbar c \cos 2\theta_{12}} L_n(r_{\text{res}}(E)), \quad (22)$$

where $L_n(r) = n_e/|dn_e/dr|$ is the local electron-density scale height and $r_{\text{res}}(E)$ is the resonance radius of section 5.3.3 below; $\gamma_{\text{res}} \gg 1$ indicates that the density varies slowly enough, relative to the local oscillation length, for the neutrino to remain in the same instantaneous matter eigenstate. At representative energies between 5 and 15 MeV, the value at the resonance layer decreases from approximately 6.4×10^3 to 1.7×10^3 . It therefore remains several orders of magnitude above the onset throughout the tested range, in agreement with the small Landau–Zener corrections of section 3.1.2. Figure 12 gives $\gamma(\rho)$ across the solar production region. Except for a narrow interval near the surface, the values remain well above unity and decrease monotonically with energy.

5.3.3 MSW resonance

The MSW resonance condition, $V_k(\rho) = \cos 2\theta_{12}$, is evaluated along the solar density profile for several neutrino energies. For $E \lesssim 1.5$ MeV, the potential does not reach this condition inside the Sun, and pp -like neutrinos propagate quasi-adiabatically without crossing the resonance. As the energy increases, the calculated resonance radius moves outward, from $\rho_{\text{res}} = 0.032$ at 2 MeV to $\rho_{\text{res}} = 0.282$ at 15 MeV. Figure 13 shows the corresponding local survival probability $P_{ee}(\rho, E)$. At low energy, it remains close to the vacuum-averaged value, $\sum_i |U_{ei}|^4 = 0.553$, throughout the Sun. At high energy, it begins near the adiabatic matter-eigenstate limit, $|U_{e2}|^2 = 0.296$, at the production point and approaches the vacuum-averaged value towards the solar surface. Both trends agree with the adiabatic MSW interpretation within the sampled energy and radial

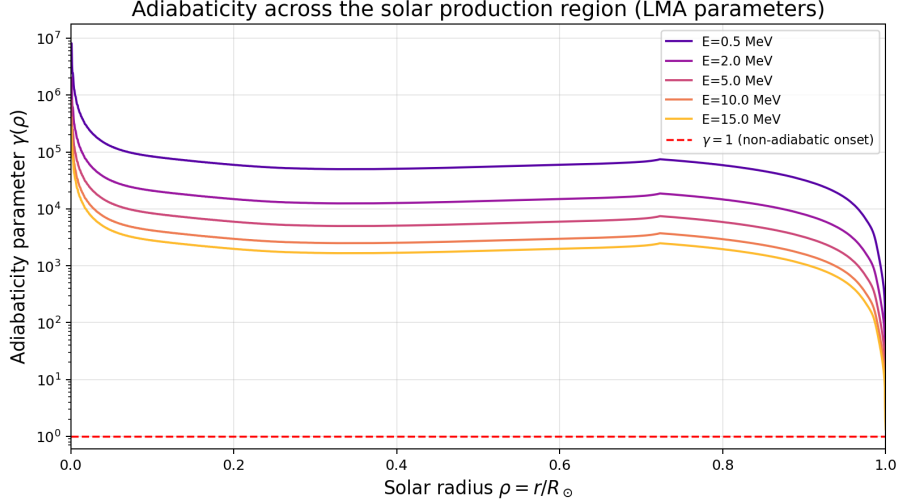


Figure 12: Adiabaticity parameter $\gamma(\rho)$ across the solar production region, for five representative energies, against the non-adiabatic onset $\gamma = 1$.

ranges.

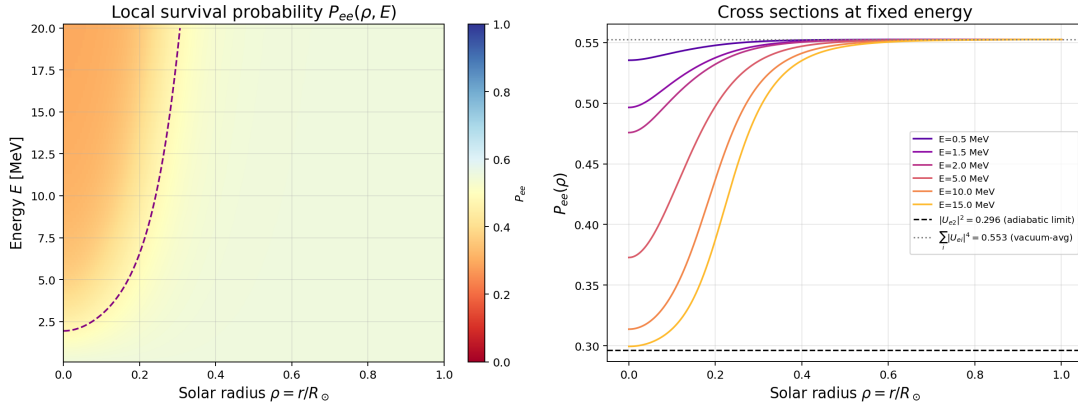


Figure 13: Local survival probability $P_{ee}(\rho, E)$ across the solar production region (left), and radial cross-sections at fixed energy (right), showing the transition between the adiabatic matter-eigenstate limit and the vacuum-averaged value.

5.3.4 CP Violation and Mass Ordering

Figure 14 shows the appearance probabilities $P_{\mu e}$ and $\bar{P}_{\mu e}$ at a long-baseline configuration ($L = 1300$ km) for the NuFIT 5.2 best-fit CP phase, $\delta_{\text{CP}} = 197^\circ$, and for maximal CP violation, $\delta_{\text{CP}} = -90^\circ$, together with the CP asymmetry $A_{\text{CP}} = P_{\mu e} - \bar{P}_{\mu e}$. The asymmetry is largest at low energy, where the genuinely CP-odd term dominates, and vanishes at the first oscillation node near $E \sim 1$ GeV, consistent with the expected L/E dependence.

The reactor-antineutrino survival probability illustrates why establishing the neutrino mass ordering is comparatively difficult. Figure 15 compares $P(\bar{\nu}_e \rightarrow \bar{\nu}_e)$ at a JUNO-like, medium-baseline configuration ($L = 50$ km) for the normal and inverted orderings: the two curves are visually almost indistinguishable, since Δm_{31}^2 and Δm_{32}^2 differ only by the small solar splitting Δm_{21}^2 , and their difference (right panel) oscillates rapidly between roughly $\pm 3\%$ rather than producing a single, easily resolved offset. Extracting the mass ordering from this observable therefore requires resolving the relative phase between the two split oscillation frequencies across a broad energy range at sub-percent energy resolution, rather than detecting one large-amplitude

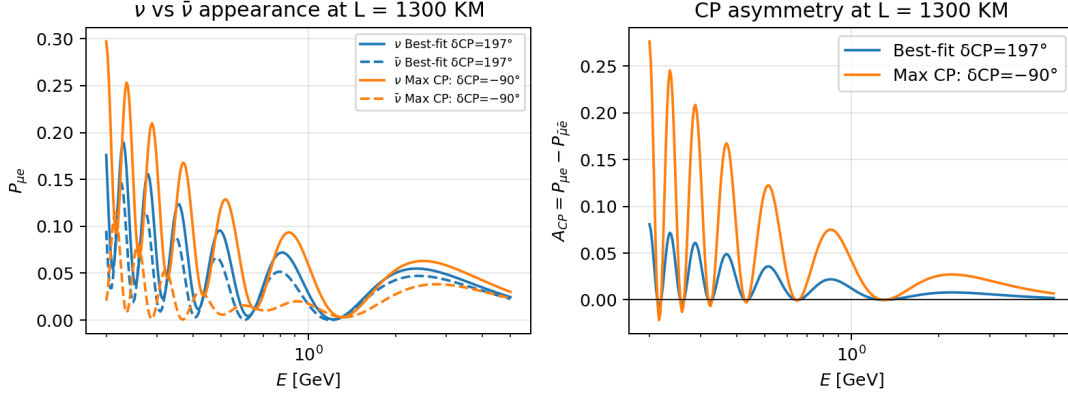


Figure 14: Appearance probabilities $P_{\mu e}$ and $\bar{P}_{\mu e}$ at a long-baseline configuration, $L = 1300$ km (left) and the resulting CP asymmetry A_{CP} (right), for the best-fit and maximal-CP-violation values of δ_{CP} .

effect, precisely the strategy pursued by medium-baseline reactor experiments such as JUNO.

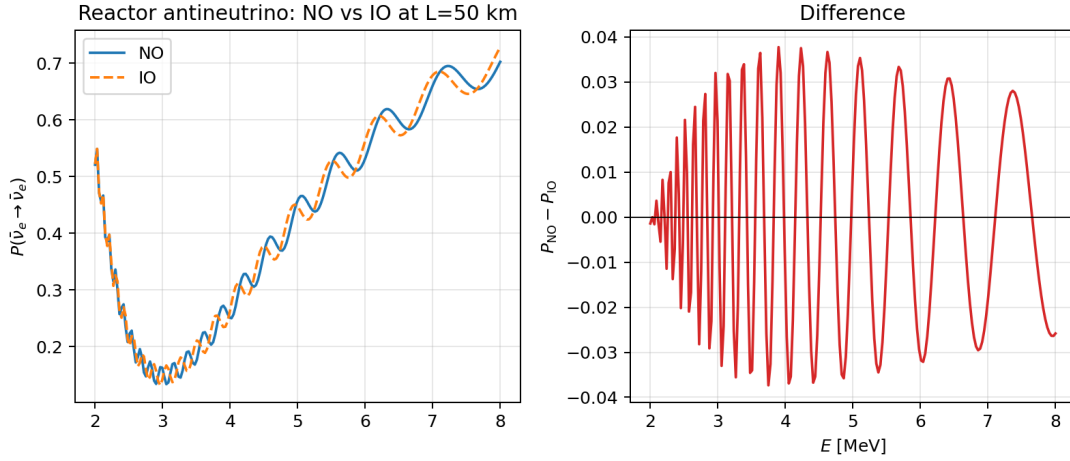


Figure 15: Reactor-antineutrino survival probability $P(\bar{\nu}_e \rightarrow \bar{\nu}_e)$ at $L = 50$ km for the normal and inverted mass orderings (left), and their difference (right).

5.3.5 Atmospheric neutrino oscillograms

Figure 16 shows the atmospheric ν_μ survival-probability oscillogram in the $(\cos\theta_z, E)$ plane, computed through the PREM Earth density profile (Dziewonski & Anderson, 1981). The characteristic vacuum oscillation dip is verified against the analytic first-minimum condition, $1.267|\Delta m_{3\ell}^2|L/E = \pi/2$ (Ashie et al., 2004), and against a set of benchmark observables summarised in table 6, spanning the atmospheric mass splitting and mixing angle, the vacuum L/E dip position, the multi-GeV up/down asymmetry, the energy of maximal mantle-crossing suppression, and the maximum appearance probability associated with the mantle MSW resonance.

5.3.6 Earth matter tomography

The derivative of the electron-neutrino appearance probability with respect to nadir angle, $|dP_e/d\eta|$, is shown in figure 17. Its localised features coincide with the PREM shell boundaries (Dziewonski & Anderson, 1981): the core–mantle density jump produces the largest spike, while mantle-only trajectories show a regular oscillatory pattern. The numerical propagation therefore retains sensitivity to the discontinuities in the density profile, which is required for the tomography applications considered here.

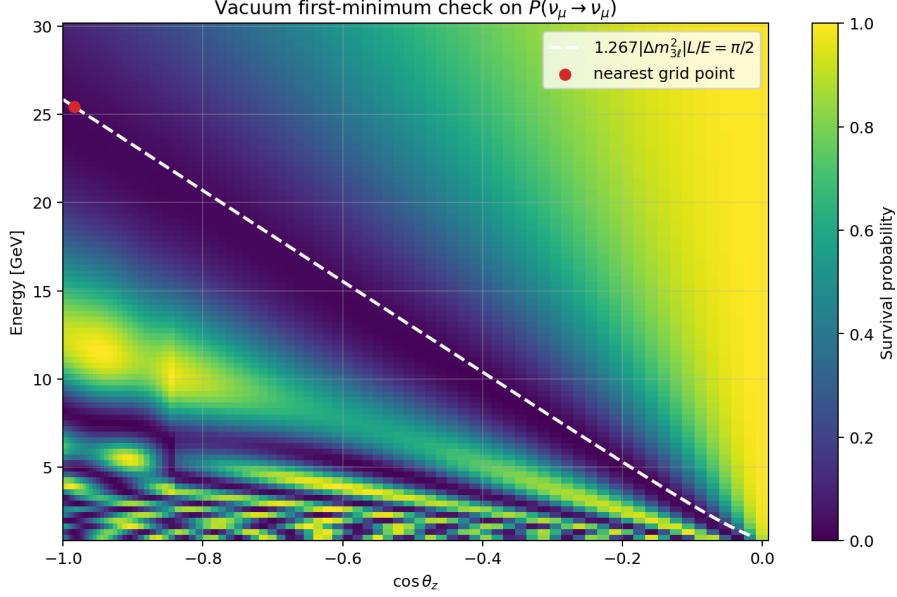


Figure 16: Atmospheric $P(\nu_\mu \rightarrow \nu_\mu)$ oscillogram in the $(\cos \theta_z, E)$ plane. The dashed line marks the analytic first vacuum-oscillation minimum, $1.267|\Delta m_{3\ell}^2|L/E = \pi/2$; the red marker shows the nearest grid point.

Table 6: Benchmark observables reproduced from the atmospheric oscillogram, compared with published experimental results.

Observable	tpeanuts	Experiment	Reference
$ \Delta m_{3\ell}^2 $ [eV ²]	2.511×10^{-3}	$[1.9, 3.0] \times 10^{-3}$	(Super-Kamiokande Collaboration, Wester, et al., 2024)
$\sin^2(2\theta_{23})$	0.981	> 0.90	(Super-Kamiokande Collaboration, Wester, et al., 2024)
L/E [km/GeV]	493.7	≈ 500	(Ashie et al., 2004)
A_μ	-0.414	≈ -0.32	(Fukuda et al., 1998)
Max-suppression [GeV]	3.28	3–30 GeV range	(IceCube Collaboration, Aartsen, et al., 2013)
$\max P(\nu_\mu \rightarrow \nu_e)$	0.5904	mantle MSW resonance	(KM3NeT Collaboration, Aiello, et al., 2024)

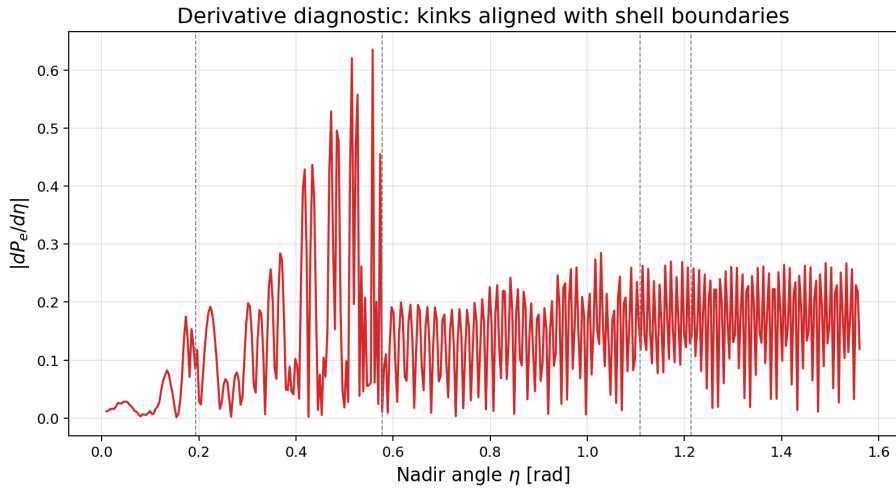


Figure 17: Derivative of the electron-neutrino appearance probability with respect to the nadir angle, $|dP_e/d\eta|$; the dashed vertical lines mark the PREM shell boundaries.

5.4 BSM extensions

5.4.1 NSI

The propagation-NSI implementation of section 2.3.1 is checked with solar ^8B presets based on literature bounds, global-fit values and projected experimental sensitivities. Figure 18 includes the Standard Model, the diagonal bound of Biggio et al. (2009), representative DUNE and Hyper-Kamiokande configurations, an IceCube-bounded $\mu\tau$ coupling, and the values of Esteban et al. (2018). All presets approach the Standard-Model vacuum-averaged and adiabatic limits at low and high energy. Differences of up to ~ 2 percentage points occur in the intermediate MSW region at $E = 10$ MeV. The global-fit and combined Hyper-K-like presets give the largest changes. The $\mu\tau$ preset is visually indistinguishable at the coupling and plotting precision used here, but this is not a general property of the NSI Hamiltonian.

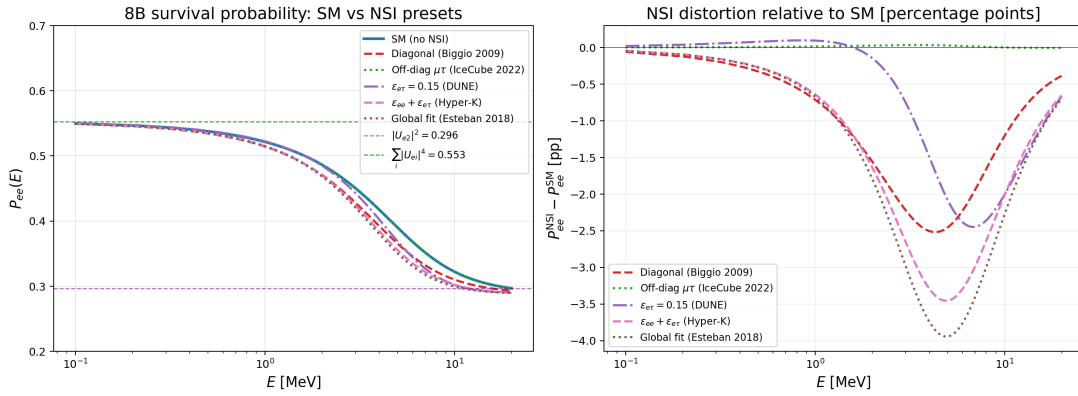


Figure 18: ^8B survival probability $P_{ee}(E)$ under the Standard Model and several representative NSI presets (left), and the resulting distortion relative to the Standard Model, in percentage points (right).

Scanning $\varepsilon_{e\mu}$ and $\varepsilon_{e\tau}$ independently from -0.3 to $+0.3$, with all remaining NSI parameters set to zero, gives the distortions shown in figure 19. Above 3 MeV, each coupling produces a monotonic change relative to the Standard Model, with opposite signs for the two scans. The calculation therefore resolves the dependence on the sign and magnitude of the selected off-diagonal coupling over the range tested.

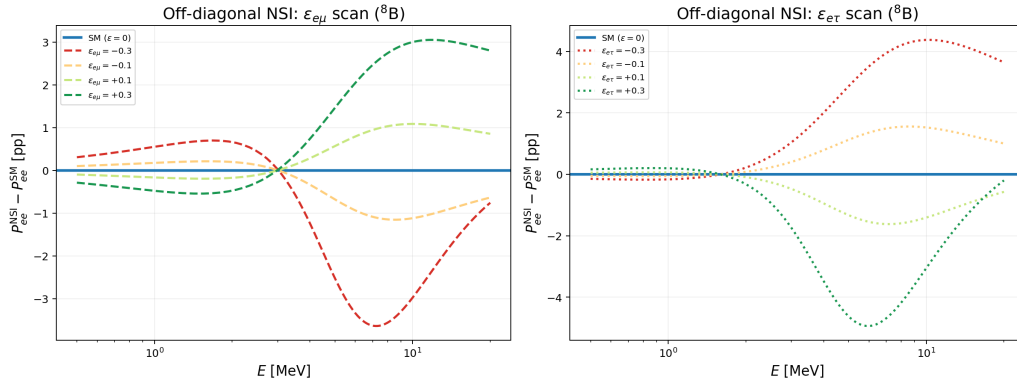


Figure 19: Distortion of the ^8B survival probability $P_{ee}(E)$ relative to the Standard Model, from independent scans of $\varepsilon_{e\mu}$ and $\varepsilon_{e\tau}$ over -0.3 to $+0.3$, with all other NSI parameters fixed to zero.

5.4.2 Sterile neutrino

The $3 + 1$ sterile extension of section 2.3.2 is checked for three vacuum presets: a null-mixing reference, a global-fit-motivated best fit (Gariazzo et al., 2017), and an IceCube benchmark

(IceCube Collaboration, Abbasi, et al., 2020). The numerical results are listed in table 7. With null mixing, the PMNS submatrix, Hamiltonian and vacuum probability recover the Standard-Model limit to floating-point precision. For all three presets, $\|U_4 U_4^\dagger - I_4\| \sim 10^{-16}$. Under the CC-only default of section 2.3.2, for which no neutron-density profile is supplied, the sterile row and column of the matter Hamiltonian vanish. A separate CP-phase scan at LSND-like kinematics gives a maximal appearance asymmetry of $|A_{\text{CP}}| \sim 10^{-5}$. This small value is consistent with the lack of simultaneous sensitivity to two independent mass-squared splittings at a single short baseline.

Table 7: Confirmed 3 + 1 sterile-extension validation results.

Check	Result
SM-limit PMNS/Hamiltonian agreement (null mixing)	exact (0 to printed precision)
Unitarity $\ U_4 U_4^\dagger - I_4\ _F$ (three presets)	$4.3\text{--}5.1 \times 10^{-16}$
Total vacuum probability $\sum_\beta P_\beta$	1.00000000, max. dev. 6.7×10^{-16}
Sterile row/column of matter Hamiltonian	exactly 0
Max. appearance CP asymmetry (LSND kinematics)	0.001% at $\delta_{14} = 108.9^\circ$

For ν_e , the two-flavour disappearance approximation agrees with the idealised limit to 10^{-14} . In the realistic-mixing case, the remaining difference is accounted for by the naive-versus-exact $|U_{e4}|^2$ convention. For ν_μ , a residual of $\mathcal{O}(10^{-2})$ remains after applying the same correction. The three presets were also evaluated at reactor, Gallium, LSND/MiniBooNE and IceCube kinematics. Combining the reactor and IceCube disappearance bounds gives $A_{e\mu}^{\text{eff}} \lesssim 0.003$, whereas the LSND region requires 0.003–0.03. This reproduces the $> 3\sigma$ appearance–disappearance tension reported for the minimal 3 + 1 model (Böser et al., 2020).

5.4.3 LMA-Dark degeneracy

Figure 20 examines the LMA-Dark transformation at fixed nadir angle, $\eta = 30^\circ$, and 1 km depth. It compares standard LMA ($\varepsilon_{ee} = 0$), an Earth-only sign-flip test ($\varepsilon_{ee} = -2$), and the simultaneous transformations $\theta_{12} \rightarrow \pi/2 - \theta_{12}$ and $\varepsilon_{ee} \rightarrow -(2 + \varepsilon_{ee})$. The respective asymmetries are -4.8% , $+1.6\%$ and $+2.9\%$. The first has the same sign as the Super-Kamiokande result, $(-3.2 \pm 1.1_{\text{stat}} \pm 0.5_{\text{syst}})\%$ (Renshaw et al., 2014), whereas the two transformed cases reverse it. For this preset and geometry, Earth regeneration therefore distinguishes solutions that are degenerate in vacuum. These are single-point diagnostics, not the flux- and exposure-averaged quantity of section 5.3.1; this accounts for the difference between -4.8% here and the integrated -2.69% prediction.

5.5 Inference with real experimental data

5.5.1 Detector-level case study: Daya Bay

The detector and inference layers of section 4.1.1 are applied to the published Daya Bay reactor-antineutrino data set following section 3.4. The input comprises 8 antineutrino detectors in three experimental halls, 6 reactor cores and the prompt-energy Inverse-Beta-Decay (IBD) spectra from the 8-detector period. The relative near/far observable provides the successful end-to-end comparison reported below. The absolute-rate prediction and the complete spectral-shape fit remain only partially validated.

At Daya Bay’s published oscillation parameters and with no free normalisation, the closure

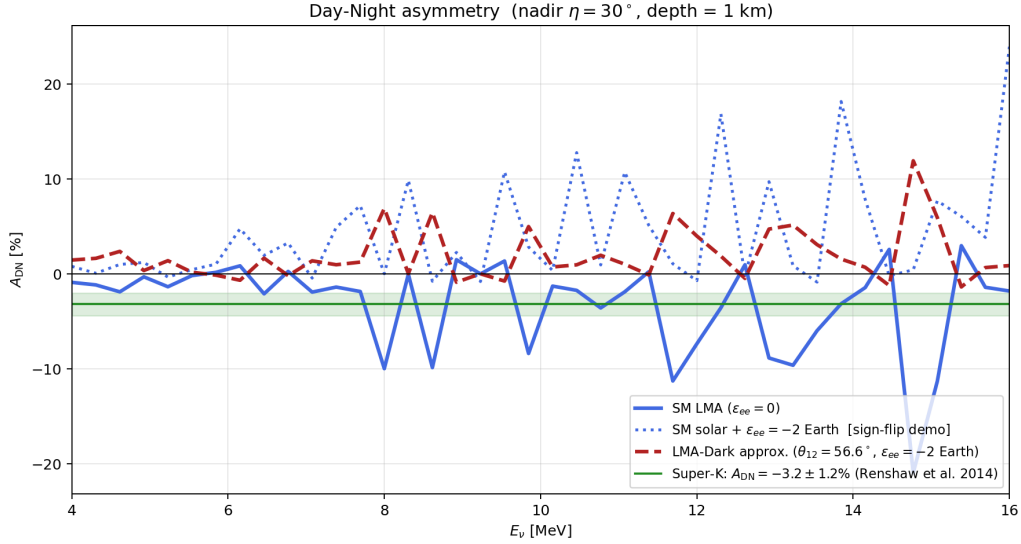


Figure 20: Day–night asymmetry versus energy at fixed nadir angle ($\eta = 30^\circ$, 1 km depth), comparing the standard LMA solution, a sign-flip demonstration with an Earth-only NSI coupling, and the full LMA-Dark approximation, against the combined Super-Kamiokande measurement (Renshaw et al., 2014).

calculation predicts 3 705 407 IBD counts across the 8 detectors, compared with 3 114 934 observed. The predicted-to-observed ratio is 1.190, with a similar excess in every detector. The normalisation audit examines the target mass, efficiency, cross section and Huber–Mueller yield. The measured-to-predicted reactor-flux ratio 0.946 ± 0.020 (An et al., 2017) changes the model ratio to $1.190 \times 0.946 \approx 1.126$. The remaining 12.6 percentage points are not identified conclusively: Daya Bay’s quoted correlated uncertainties, including approximately 2% for the combined detection efficiency, are too small to explain them alone. This residual is therefore retained as an unresolved component of the absolute-rate calculation. The fits below include a free ν_{norm} nuisance parameter to absorb this normalisation difference.

The predicted and observed prompt-energy spectra have the same IBD-kinematic structure for the selected near- and far-hall detectors. Relative to the no-oscillation expectation, the far-hall spectrum shows the expected suppression. Three fits of the complete 8-detector, 208-bin spectrum over $(\theta_{13}, \Delta m_{31}^2, \nu_{\text{norm}})$ were attempted with progressively more nuisance freedom in the background rates and energy-scale non-linearity. None converged to a physical point. With the energy-scale pulls, $\hat{\theta}_{13}$ moves to 0° , indicating a degeneracy between the oscillation deficit and the uncorrelated energy-scale parameters. The full spectral fit is consequently retained as a limitation rather than used as a physics result.

The two reduced observables do converge. When each detector spectrum is integrated to one rate, fitting $(\theta_{13}, \nu_{\text{norm}})$ with fixed Δm_{31}^2 gives $\hat{\theta}_{13} = 8.389^\circ \pm 0.139^\circ$, or $\sin^2 2\theta_{13} = 0.08332$. This is -2.66% relative to Daya Bay’s published 0.08560, corresponding to -0.85σ . The near/far ratio fit follows the strategy of the 2012 Daya Bay analysis (An et al., 2012) and is the only calculation here with joint sensitivity to both oscillation parameters:

The two-dimensional $-2 \ln L$ scan in figure 21 places the fitted point and Daya Bay’s published point in the same low- $\Delta(-2 \ln L)$ region. In table 8, the jointly fitted $\sin^2 2\theta_{13}$ and Δm_{31}^2 differ by 1–2% from the Daya Bay values and lie inside the NuFIT global-fit range. This calculation uses the multi-reactor flux, order- $1/M$ cross section and IAV+LSNL+Gaussian response implemented

in **tpeanuts**, together with the experimental rather than synthetic data set. It exercises the detector and inference layers on real data; its scope is the relative near/far observable, subject to the absolute-rate and spectral-shape limitations described above.

Table 8: Daya Bay near/far ratio fit vs. Daya Bay’s own published best fit and NuFIT 6.1 (Esteban et al., 2024, 2025).

Quantity	tpeanuts (near/far)	Daya Bay published	NuFIT 6.1
$\sin^2 2\theta_{13}$	0.08442	0.0856 (An et al., 2023)	0.0878
θ_{13} [deg]	8.445 ± 0.134	8.506	8.62
Δm_{31}^2 [10^{-3}eV^2]	2.478 ± 0.055	2.528	2.511

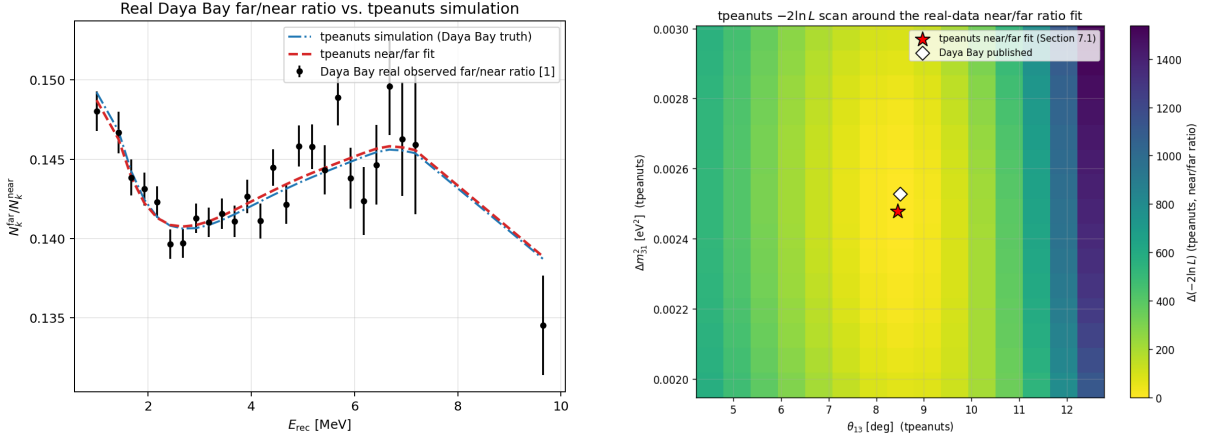


Figure 21: Real Daya Bay far/near per-bin count ratio, compared against the **tpeanuts** simulation at Daya Bay’s own published truth and at the near/far fit (left); **tpeanuts** $-2\ln L$ scan around that fit, in the $(\theta_{13}, \Delta m_{31}^2)$ plane, compared with Daya Bay’s own published best fit (right).

5.5.2 Other detector-inference applications

- **IceCube DeepCore:** a 4-parameter Poisson fit to 9 years of golden atmospheric-event data recovers $\theta_{23} = 45.42^\circ \pm 1.85^\circ$ and $\Delta m_{3l}^2 = (2.395 \pm 0.039) \times 10^{-3} \text{eV}^2$, a -0.05σ pull on $\sin^2 \theta_{23}$ relative to the collaboration’s published best fit; a dedicated audit of the fitted muon-background normalisation (16% above the published value) rules out an Earth-matter-unitarity origin (a 0.002% effect) and instead suggests most of the gap to fixed detector-systematics hypersurfaces (up to +12% for the ν_τ -CC channel) not yet freed as nuisance parameters.
- **SNO phase II:** the fit uses the full 391-day, 38-observable salt-phase data set, the published statistical covariance and a reconstructed systematic covariance that does not yet include the dedicated NC-channel systematics. It gives $\theta_{12} = 35.28^\circ \pm 1.61^\circ$ and $\Delta m_{21}^2 = (6.10 \pm 34,900) \times 10^{-5} \text{eV}^2$. The value of θ_{12} lies within SNO’s SNO-only 1σ range. The very large uncertainty on Δm_{21}^2 indicates that the local Laplace approximation breaks down when the highly correlated reconstructed covariance is included; a likelihood scan, as implemented in section 4.1.1, is more appropriate in this case.
- **Borexino:** a direct fit of θ_{12} to the four published $P_{ee}(E)$ points (Borexino Collaboration, Agostini, et al., 2018), starting from a deliberately uninformative guess (maximal mixing), converges to $\theta_{12} = 34.59^\circ \pm 4.15^\circ$, within 0.2σ of NuFIT. The separate, full detector-level elastic-scattering event-rate chain (target, response, cross section) is implemented and unit-tested against the real Nature 2018 low-energy spectrum. But, its background model

is currently a documented placeholder, so no event-rate fit to real data is yet reported. A joint two-parameter extension, freeing a diagonal Non-Standard-Interaction coupling ε_{ee} alongside θ_{12} , recovers a result compatible with the real LMA-Dark degeneracy (Miranda et al., 2006) directly from the same four points. Two independently-converged fits started on opposite sides of $\theta_{12} = 45^\circ$ land on statistically indistinguishable χ^2 (0.737 each) at well-separated $(\theta_{12}, \varepsilon_{ee})$, related by the expected $\theta_{12} \rightarrow 90^\circ - \theta_{12}$, $\varepsilon_{ee} \rightarrow -(2 + \varepsilon_{ee})$ transformation.

- **SNO phase I:** a day/night spectral fit to real published data converges robustly to $\theta_{12} = 34.333^\circ \pm 0.665^\circ$, a 0.86σ pull relative to NuFIT 6.1.

Table 9 summarises these four results against their own published references.

Table 9: `tpeanuts` inference results vs. published values for the four experiments above.

Experiment	Quantity	<code>tpeanuts</code>		Published	Diff. (%)	Reference
IceCube DeepCore	θ_{23} [deg]	45.42 ± 1.85	45.57 ($\sin^2 \theta_{23} = 0.51$)		-0.34	(Abbasi et al., 2023)
IceCube DeepCore	Δm_{3l}^2 [10^{-3} eV^2]	2.395 ± 0.039		2.484	-3.59	(Abbasi et al., 2023)
SNO II	θ_{12} [deg]	35.28 ± 1.61	33.85 ($\tan^2 \theta_{12} = 0.45$)		+4.21	(Aharmim et al., 2005)
SNO II [†]	Δm_{21}^2 [10^{-5} eV^2]	$6.10 \pm 34,900$		$5.0^{+0.2}_{-1.8}$	+22.0	(Aharmim et al., 2005)
Borexino*	θ_{12} [deg]	34.59 ± 4.15		33.76 (NuFIT)	+2.47	(Esteban et al., 2025)
SNO I	θ_{12} [deg]	34.333 ± 0.665		33.76 (NuFIT)	+1.70	(Esteban et al., 2025)

*Direct fit of $P_{ee}(E)$ to Borexino’s own published points, not the full detector-level event-rate fit, which is still pending (see text above). [†]The quoted Diff. is not statistically meaningful: the fitted uncertainty itself is $\sim 5700\%$ of the central value, reflecting the Laplace approximation’s breakdown for this parameter (see text above).

5.6 Public repository and supplementary material

The full source code of `tpeanuts`, together with the notebooks used for validation, analysis and benchmarking, is publicly available in the following GitHub repository:

<https://github.com/juanramondiaz/tpeanuts>

The repository is released under the GNU General Public License v3 (GPLv3), the same licence as the original PEANUTS implementation it builds on. The repository is organised so that the physical modules are kept separate from the notebooks that exercise them; `notebooks/` is the main entry point for reviewing results and reproducing figures (table 10).

Full architectural and physics detail beyond the scope of this thesis, module by module, is documented separately in seven companion L^AT_EX documents shipped alongside the code (`documentation/`), each with its own figures and, for the inference documents, its own real-data results: `guide` (`core`, `medium`, `pipeline`, `detector.common/interaction/borexino`); `BSM_NSI` and `BSM_sterile` (theory, implementation and notebook results for each BSM extension, table 7); and `inference_dayabay`, `inference_icecube`, `inference_sno_ii`, `inference_borexino` (full detector implementation and real-data results underlying sections 5.5.1 and 5.5.2).

Table 10: Notebook directory tree (paths relative to `notebooks/` in the repository above).

Path	Contents
<code>physics/simulation/{vacuum,solar,earth,atmosphere}</code>	Fundamentals, production/exposure profiles and differential-flux analysis per medium.
<code>physics/experimental_results/{vacuum,solarneutrino,atmosphericneutrino,BSM}</code>	Reproduction of the physical signatures of section 5.3 and table 7.
<code>validation/{nusquids,legacy}</code>	Cross-validation against <code>nuSQuIDS</code> and bit-level validation against legacy PEANUTS.
<code>inference/</code>	Detector-level closure tests and gradient-based fits to real data.
<code>benchmark/</code>	Performance comparison against PEANUTS and <code>nuSQuIDS</code> .
<code>external/{corsika,nusquids,prem,zenodo,reactor/dayabay}</code>	CORSIKA generation and exploratory/data-retrieval notebooks for external inputs.

6 Conclusions

This thesis has described the motivation, theoretical basis and development of `tpeanuts`, a public modular PyTorch framework for neutrino-oscillation phenomenology derived from the PEANUTS toolkit. In the translated part of the code, the PEANUTS physical conventions are retained while its numerical operations are adapted to tensors, batched evaluation and accelerator execution. The remaining modules extend this translated implementation.

The translated numerical core covers PMNS construction, matter potentials, vacuum propagation, matter mixing and solar-neutrino probabilities. Its comparison with PEANUTS includes low-level numerical quantities, intermediate physical outputs and final probabilities. These tests define the numerical basis on which the additional components are constructed.

The main contribution is the modular architecture, which separates the oscillation engine from the neutrino source, matter-density model and physical scenario. Starting from the solar-neutrino scope of PEANUTS, the framework adds atmospheric-neutrino detection through the atmosphere and Earth, interchangeable solar and terrestrial density models, perturbative treatments of the matter potential and Beyond the Standard Model extensions (non-standard interactions and sterile neutrinos). These additions use the same propagation interface.

Another important contribution is the detector and statistical-inference framework. A detector-independent forward-folding layer and a gradient-based likelihood layer use PyTorch autograd to connect the oscillation calculation to published data from four experiments, or five detector data sets when the two SNO phases are counted separately. In the Daya Bay case study, a relative near/far fit recovers both oscillation parameters within 1–2% of the published values. The same study leaves an approximately 12.6-percentage-point residual in the absolute-rate prediction and does not obtain a convergent full-spectral-shape fit without a complete systematic covariance. The SNO II result also shows the breakdown of the local Laplace approximation for Δm_{21}^2 . These limitations define specific parts of the inference framework that still require further work.

The propagation, detector and inference components are distributed together as a documented, installable Python package, publicly released on GitHub under the GNU General Public License v3 (GPLv3). Its modular interfaces allow the calculations to be reused or extended for solar, atmospheric and BSM neutrino studies.

Future work includes improving the correlated-systematic treatment in the detector analyses

and connecting the sterile-neutrino extension to the real-data inference machinery used for the Standard-Model detector layer. The current Majorana-neutrino parametrisation can be extended with magnetic-field effects, a transition magnetic moment and the corresponding spin-flavour precession Hamiltonian, through which Majorana and Dirac neutrinos become experimentally distinguishable. Further developments include propagation through arbitrary density profiles and the combination of the individual detector-inference calculations into a joint likelihood across experiments.

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